

# DIVE DEEPER, BRANCH LATER: SELF-EVOLVING EVALUATORS ACROSS OPEN-ENDED TASKS

**Anonymous authors**

Paper under double-blind review

*Working manuscript: results are placeholders or simulated illustrations; empirical evaluation pending.*

## ABSTRACT

Evaluating open-ended outputs requires task-dependent judgments over multiple aspects of quality that must be balanced in context. Existing proxy metrics and model-based evaluators remain imperfect on such judgments and do not consistently align with human preferences. Humans can often tell which of two concrete outputs is better, but it is much harder to turn that ability into a comprehensive and effective evaluation procedure. We present IterEval, a framework for self-evolving evaluators across open-ended tasks. IterEval selects human preferences through disagreement-based preference mining and uses them to iteratively improve the evaluation procedure. An evaluation error revealed by human preferences often suggests several plausible ways to improve the evaluator, while limited human feedback provides only weak evidence for deciding which direction to pursue. This ambiguity matters especially in open-ended evaluation, where a useful direction may require several related revisions before producing a measurable gain. An unsuccessful early candidate evaluator therefore leaves the optimizer uncertain whether the direction is wrong or simply underdeveloped. In the default evaluator-evolution setup we tested, this uncertainty often led subsequent revisions toward alternative directions rather than deeper development of the current one. Depth-first evaluator evolution explicitly maintains an improvement direction across revisions and continues developing it until diagnostic results provide a clear reason to move on or its search budget is exhausted. The search keeps candidate acceptance separate from direction continuation, allowing diagnostic results from unsuccessful candidates to guide further implementation without replacing the best evaluator found so far. Across diverse open-ended tasks, our qualitative analyses show that IterEval develops promising directions more thoroughly, produces more sustained improvements, and yields evaluators that better reflect human preferences under matched search budgets. These improvements also transfer downstream: evolved evaluators support better output selection and more effective critic-guided refinement. On representative tasks, using their scores as training rewards further improves generation quality.

## 1 INTRODUCTION

Evaluating open-ended tasks requires case-specific judgments across multiple dimensions of quality, with no single evaluation standard sufficient for every case (Ye et al., 2024; Kim et al., 2024b). A summary may need to preserve fine-grained details when completeness matters, yet emphasize brevity and the central argument when rapid understanding is the goal. Likewise, visual richness may be valuable for a promotional webpage but detrimental to a task-oriented interface. Metrics and model-based evaluators capture useful aspects of quality, yet do not consistently match human preferences (Fabbri et al., 2021; Xu et al., 2023; Zheng et al., 2023; Hu et al., 2024). Humans themselves, however, often cannot fully express their judgments as a comprehensive, executable evaluation procedure (Shankar et al., 2024b). Human preferences are therefore better treated as iterative feedback for improving an evaluator.

Existing approaches use human feedback to train evaluator models (Xu et al., 2023), learn or refine evaluation criteria (Liu et al., 2024; Lv et al., 2026), and select evaluation functions (Shankar et al.,

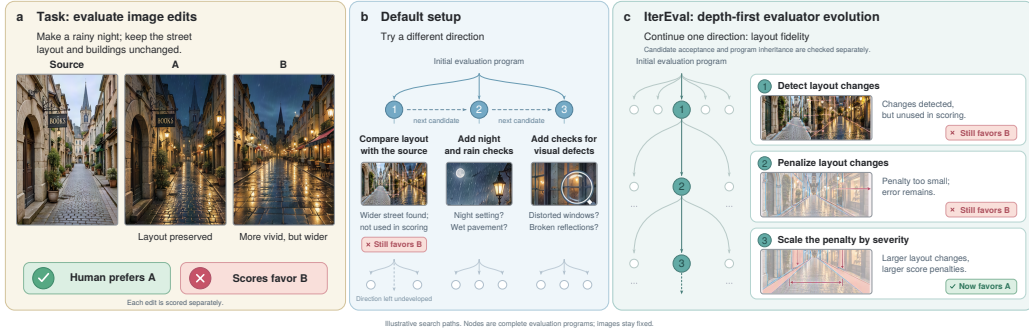


Figure 1: Dive deeper, branch later. Panel (a) contrasts layout fidelity with visual richness. Panel (b) branches across plausible corrections; dashed arrows show attempt order. Panel (c) develops one correction through detection, scoring, and severity weighting. Nodes denote complete evaluation programs; hollow nodes are unexplored alternatives. The corrected ordering is local to this example, not candidate acceptance or generalization. Images, preferences, and paths are constructed illustrations, not experimental evidence.

2024b). Prompt and skill optimizers offer a complementary mechanism for revising instructions from execution feedback (Agrawal et al., 2026; Yang et al., 2026). We present IterEval, a framework for self-evolving evaluators across open-ended tasks. IterEval directly revises an explicit evaluation procedure while building on the base model’s existing semantic and reasoning capabilities. This allows the evaluator to adapt without updating the base model parameters and makes individual changes easier to inspect and test.

Extending evaluator self-evolution to new open-ended domains, however, raises two challenges: obtaining useful human preferences and reliably turning them into improvements to the evaluation procedure.

Obtaining preference data that is useful for evolution is itself nontrivial: under a limited human feedback budget, the choice of comparisons can affect learning efficiency (Muldrew et al., 2024; Oh et al., 2025). Hand-designed comparisons tend to reflect the cases and failure modes anticipated by the designer, while broad collections can contain many easy or redundant comparisons that reveal little about where the evaluator still falls short. We therefore use disagreement-based preference mining to prioritize comparisons on which existing evaluation signals disagree. When preference labels already exist, preference mining selects informative examples from the available data; otherwise, it prioritizes comparisons for new human annotation.

Preference mining selects the comparisons used for evaluator evolution, but does not specify how the evaluator should be revised. Pairwise preferences identify the better output without specifying the fine-grained criteria behind that choice (Ye et al., 2024; Lv et al., 2026). The same evaluation error revealed by human preferences can therefore suggest several plausible improvement directions, while limited human feedback often provides little basis for choosing among them. This ambiguity is especially important in open-ended evaluation, where a useful direction may require several related revisions before producing a measurable gain. An unsuccessful early candidate evaluator therefore leaves open whether its direction is ineffective or simply underdeveloped.

This uncertainty also affects how evaluator evolution proceeds. In the default evaluator-evolution setup we tested, each new revision was proposed by the model from previous candidates and their outcomes. We observed a tendency for unsuccessful early candidates to be followed by revisions in alternative directions rather than deeper development of the current one. Over repeated iterations, evolution can therefore spread effort across multiple plausible directions before any one of them has been developed far enough to judge reliably.

IterEval uses depth-first evaluator evolution to explicitly maintain an improvement direction across revisions (Figure 1c; Figure 2c). The search continues developing the current direction after unsuccessful intermediate candidates, using their diagnostic results to guide the next revision. The best evaluator is maintained separately and replaced only by an improved candidate. Evolution moves to

a new direction when diagnostic results provide sufficient evidence against the current one or when its search budget is exhausted.

We evaluate IterEval across structured visual artifacts, images and 3D, text generation, and automated research, covering both reused preference datasets and newly collected human preferences. Disagreement-based preference mining identifies preference subsets that lead to stronger evaluator improvement than standard sampling from the same data source. Under matched search budgets, depth-first evaluator evolution continues developing directions after unsuccessful early candidates and produces more sustained improvement than the default setup. The resulting evaluators also achieve stronger agreement with held-out human preferences and improve downstream output selection and critic-guided refinement. On representative tasks, their scores further provide useful training rewards for improving generation quality.

Our contributions are as follows:

- We introduce IterEval, a framework that learns from human preferences and revises the full evaluation procedure across heterogeneous open-ended task domains.
- Depth-first evaluator evolution explicitly maintains an improvement direction across revisions, preventing an unsuccessful intermediate candidate from automatically ending development of that direction. Disagreement-based preference mining selects informative preferences from existing labeled data or prioritizes comparisons for new human annotation.
- We provide a cross-domain evaluation of preference mining, evaluator evolution, final agreement with human preferences, and downstream utility under controlled annotation and search budgets.

## 2 RELATED WORK

**Learning evaluation procedures from human feedback.** Reward hacking exposes a limitation of improving generators against an imperfect evaluator: optimization can increase evaluator scores by exploiting errors in how quality is assessed (Wang et al., 2026). This motivates using human feedback to improve the evaluation procedure itself, rather than treating it as a fixed objective. We share the goal of aligning judgments with human preferences with learned reward models and trained judges (Xu et al., 2023; Kim et al., 2024a; Ke et al., 2024). Evaluator fine-tuning changes model weights, whereas our search changes an executable evaluation program with the base model fixed. Criteria-learning methods, including preference-grounded query-specific rubrics, are closer to our approach because they use human feedback to develop explicit evaluation rules (Shankar et al., 2024a; Liu et al., 2024; Shankar et al., 2024b; Lv et al., 2026). Our search also changes how evidence is gathered and how it affects the final judgment, allowing a correction to span several steps of the evaluation procedure. Tool-based inspection and evolving verifier skills already appear in Agent-as-a-Judge and WebGrader (Zhuge et al., 2025; Chen et al., 2026a); our question is how to adapt that procedure to human quality trade-offs that execution checks alone do not specify.

**Continuing directions in program evolution.** Our depth-first evaluator evolution builds on methods for revising prompts, modules, and agent code (Khatab et al., 2024; Yuksekgonul et al., 2025; Agrawal et al., 2026; Hu et al., 2025; Zhang et al., 2026). Its focus is how search proceeds after a candidate fails to improve the score. SkillOpt already uses a rejected-edit buffer to inform later proposals while accepting only score-improving edits (Yang et al., 2026). Our search makes a further distinction between retaining that history, continuing the same direction, and inheriting the rejected implementation. An eligible candidate can remain the working program without becoming the best program, and a direction can remain active even when inheritance fails.

Persistent search is likewise not unique to IterEval: HarnessBank maintains distinct harnesses, and SkillHEX uses executable hypothesis tests and a persistent revision tree (Luo et al., 2026; Feng et al., 2026). SkillHEX addresses similarly ambiguous failures through test-time branch selection under sparse terminal rewards, without training or validation sets. Our depth-first evaluator evolution instead allocates successive revisions to an active direction, subject to diagnostic and budget checks, while measuring complete evaluation programs against a development cohort of human preferences. SkillLens’s distinction between extracting a skill and successfully consuming it reinforces the need

to examine executed behavior, not merely the plausibility of a proposed revision (Huang et al., 2026). Our comparison tests whether depth-first direction continuation improves evaluator alignment over alternative search allocations under matched budgets.

**Mining preferences for evaluator evolution.** Disagreement-based preference mining follows active preference learning in allocating limited human feedback selectively, with disagreement-based sampling rooted in query-by-committee (Seung et al., 1992; Muldrew et al., 2024; Oh et al., 2025). It applies this principle to evaluator evolution by reducing the priority of repeated comparisons on which existing evaluation signals agree, while retaining a random anchor for shared blind spots. Unlike Benchmark<sup>2</sup>, which selects test sets to preserve ranking consistency, discriminability, and capability alignment (Qian et al., 2026), our preference mining selects comparisons to improve the evaluator rather than measure competing models. Existing datasets can supply both comparisons and reusable preferences; new annotation fills missing feedback. The relevant endpoint is the resulting evaluator’s agreement at matched human cost, rather than the selected set’s disagreement or discriminability.

### 3 SCOPE AND PROBLEM FORMULATION

A *domain* is a collection of tasks sharing core quality dimensions, their priorities, the overall decision rule, and the human annotation protocol. An artifact type does not define a domain: screenshot-conditioned webpage reconstruction and specification-conditioned webpage generation require different standards. A change of source distribution that preserves the standard is instead a within-domain distribution shift. Appendix A enumerates the full research scope of 22 domains in four families, without assuming that all are ready for experiments. Closed tasks whose complete quality criterion is an executable postcondition fall outside this scope.

Existing human preferences are not a domain-admission requirement. Their availability determines the missing data work, not the algorithm’s definition. An evaluation ecosystem can exist without a suitable preference dataset. We distinguish *reused preferences*, which test improvement on established benchmarks, from *new preferences*, which test applicability when matching human judgments must first be collected. One domain can support both settings.

Let  $x$  be a task with permitted context,  $y$  a candidate output, and  $\mathcal{Q}_D = \{q_1, \dots, q_m\}$  the quality dimensions for domain  $D$ . A human comparison has the core form

$$h = (x, y_a, y_b, q, p, a, \nu), \quad (1)$$

where  $q$  is a dimension or overall quality,  $p$  is a preference, tie, or abstention,  $a$  records annotator agreement, and  $\nu$  records provenance. Human feedback has four levels:  $H_0$  gives overall preferences;  $H_1$  gives dimensional preferences;  $H_2$  identifies error type, severity, and location;  $H_3$  optionally supplies a free-form rationale.  $H_1$  is central to the dimension-guided design. An  $H_0$ -only corpus remains usable, but it supplies neither human dimensional judgments nor human explanations.

The learned object is an executable evaluation program

$$E_\theta = (A, K, V, T, \Pi, G), \quad (s_\theta, \tau_\theta) = E_\theta(x, y), \quad (2)$$

comprising artifact adapters  $A$ , dimension schema and rubrics  $K$ , evaluator modules  $V$ , tools  $T$ , evidence routing  $\Pi$ , and fusion  $G$ . The score  $s_\theta$  is accompanied by an execution trace  $\tau_\theta$ . The base model is fixed outside  $\theta$ . Self-evolution means automated changes to this evaluation program under external human feedback, not self-definition of the human standard. At deployment, the evaluator receives one candidate output and its permitted evidence; opponents, human labels, generator identity, split labels, and mining strata are excluded. We seek agreement with unseen human preferences at an explicitly reported evaluation cost, rather than a single judge shared across all domains.

### 4 DISAGREEMENT-BASED PREFERENCE MINING

Disagreement-based preference mining lowers the sampling priority of comparisons on which existing evaluation signals consistently agree. Repeated annotations of these comparisons may add

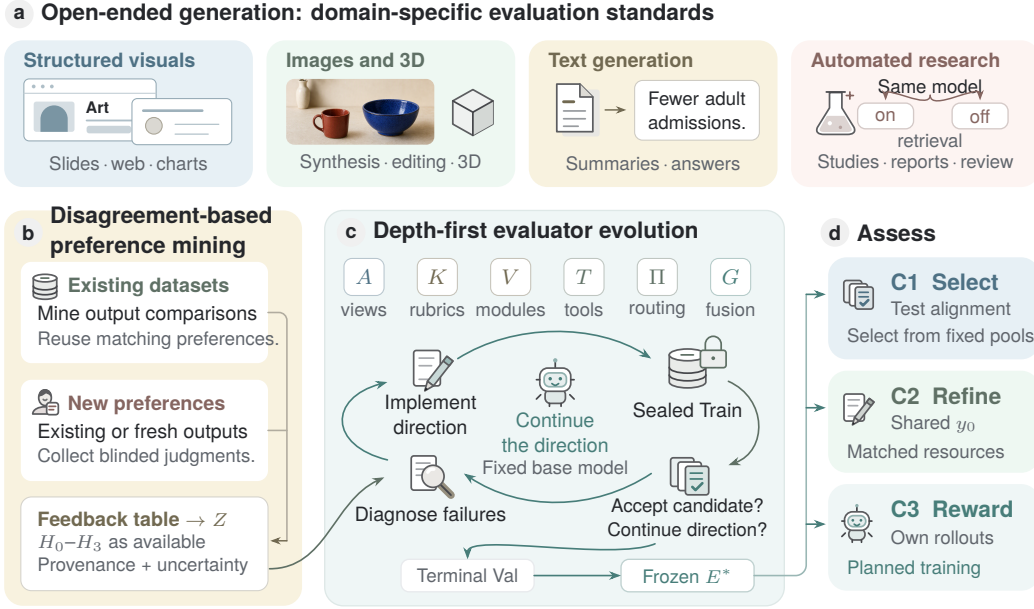


Figure 2: IterEval: iterative evaluator self-improvement from human feedback. Panel (a) illustrates four artifact families with distinct evaluation standards. Panel (b) mines comparisons from existing datasets or fresh outputs, reusing compatible preferences or collecting missing ones. Panel (c) revises a six-component evaluation procedure with a fixed base model. The loop can continue a direction after an unaccepted candidate; candidate acceptance, program inheritance, and direction continuation remain separate decisions. Search can switch directions when the current direction pauses or closes. Terminal validation does not trigger edits or reranking. Panel (d) covers C1 alignment and output selection, C2 critic-guided refinement, and planned C3 reward-guided training. Artifacts are constructed illustrations; the photograph is generated with GPT Image 2. Arrows denote information flow, not measured improvements.

little information about evaluation errors. IterEval selects comparisons from existing datasets or newly generated outputs, reusing compatible human preferences and collecting missing ones while retaining a random anchor.

#### 4.1 COMPARABLE EVALUATION SIGNALS

Outputs are grouped by their original task input and deduplicated before comparison. For a pair  $r = (x, y_a, y_b)$ , signal  $j$  supplies a preference sign  $u_j(r) \in \{-1, 0, +1, \perp\}$  from its own score difference and declared tie margin. Zero denotes a tie and  $\perp$  missing or abstaining output; raw scores from different signals are never subtracted. Signals require executable sources with compatible inputs; provenance, repeated-run variation, and missingness are recorded separately.

We freeze a many-to-many mapping  $M_{jq}$  from signal  $j$  to dimension  $q$  using its definition and measurement procedure before reading preference labels. A mixed score may connect to several dimensions, but remains one shared signal. Signals sharing a model, training labels, or underlying metric are assigned source groups so that duplicated prompts do not become independent votes. Metrics reserved for final independent assessment remain outside mining, evaluator tools, critique, and reward.

#### 4.2 SAMPLING PRIORITY

The ranking distinguishes three cases. *Within-dimension disagreement* occurs when signals from distinct source groups favor opposite outputs on the same criterion. *Overall-dimension disagreement* occurs when an overall signal opposes a dimensional signal. *Cross-dimensional trade-offs* occur when dimension-level preferences favor different outputs; these are not contradictory measure-

ments of one criterion. Dimensional preferences aggregate at most one consensus sign per source group, with internally opposed groups abstaining. Writing their indicators as  $I_{\text{within},q}$ ,  $I_{\text{overall},q}$ , and  $I_{\text{tradeoff}}$ , the implemented priority is

$$P(r) = 2 \sum_q I_{\text{within},q}(r) + \sum_q I_{\text{overall},q}(r) + I_{\text{tradeoff}}(r). \quad (3)$$

Pairs are ranked by decreasing  $P(r)$ . This structural score does not estimate information gain or human reliability. In particular,  $P(r) = 0$  can reflect missing signals or ties rather than agreement, and agreement itself need not be correct.

The random anchor precedes ranking, allowing checks for shared blind spots even among low-priority comparisons. Anchor annotations count toward the same human budget as targeted annotations. Human-reliability qualification remains separate from the implemented ranking (Appendix B). Only evolution data undergo preference mining; final evaluation uses independently specified random or stratified sampling. Whether preference mining produces more informative preferences is tested by subsequent evaluator improvement at equal human cost, not by the number of disagreements collected.

#### 4.3 HUMAN FEEDBACK FOR EVOLUTION

Existing preferences can be reused when the task, readable candidate outputs, and judged quality construct match. Derived comparisons retain their ranking, rating, or error-annotation provenance. When matching preferences are absent, the task and annotation protocol are fixed before collecting blinded judgments on existing or freshly generated outputs.

The feedback table  $Z(r, q)$  links artifact provenance, available human judgments, source signals, evaluator decisions, tool observations, and execution cost. Both construction settings supply this interface to depth-first evaluator evolution, while retaining their label origins and annotation protocols. Overall and dimensional preferences reveal evaluation errors for the search to investigate, with localized diagnostics and human rationales included when available. Missing feedback levels remain absent, and model-generated explanations are not human  $H_1$  or  $H_2$ .

### 5 DEPTH-FIRST EVALUATOR EVOLUTION

IterEval uses depth-first evaluator evolution to search over executable evaluation programs with a fixed base model, using errors revealed by human feedback. Its central distinction is between candidate acceptance and direction continuation: an unsuccessful candidate need not replace the best evaluator for its direction to receive further work. Program inheritance separately determines which implementation that work starts from.

#### 5.1 OBJECTIVE AND MEASUREMENT

For domain  $D$ , let  $\mathcal{P}_D$  sample an input and then uniformly sample a strict human preference within its output group. With a fixed repeated-scoring protocol, the target is

$$J_D(\theta) = \Pr_{(x, y^+, y^-) \sim \mathcal{P}_D, \xi} [\bar{s}_\theta(x, y^+) > \bar{s}_\theta(x, y^-)], \quad (4)$$

where  $\xi$  includes randomness in the  $R$  executions used for each median score. Each execution is pointwise (Equation 2), without access to the opponent or preference label. The default synthesizer receives specialist findings, reasons, and scopes, but not structured scores or scales (Figure 4; Appendix E).

Search estimates agreement on a finite cohort, requesting  $R \geq 3$  executions per artifact for full-evaluator measurements. For complete executions on input groups  $\mathcal{S}$  with nonempty strict-preference sets  $P_g$ , define

$$\begin{aligned} \bar{s}_\theta(x, y) &= \text{median}_{r=1, \dots, R} s_\theta^{(r)}(x, y), \quad z_{gj} = \mathbb{I}[\bar{s}_\theta(x_g, y_{gj}^+) > \bar{s}_\theta(x_g, y_{gj}^-)], \\ \hat{J}_\mathcal{S}(\theta) &= \frac{1}{|\mathcal{S}|} \sum_{g \in \mathcal{S}} \frac{1}{|P_g|} \sum_{j \in P_g} z_{gj}. \end{aligned} \quad (5)$$

Input groups receive equal weight; predicted ties count as incorrect. The native development proxy uses one fixed pair per input and need not represent every within-group preference. Missing executions retain their denominators through coverage and missing-outcome bounds (Appendix F).

## 5.2 DIRECTIONS AND CANDIDATES

A direction  $d$  specifies a correction to the evaluation procedure, motivated by a hypothesis about how it would repair an observed error. Implementing that correction may require coordinated changes across  $(A, K, V, T, \Pi, G)$ . For example, correcting omission judgments can require finding missing content, assessing its importance, and making that assessment affect the score. Changing only one step may leave the error unresolved. A candidate  $C$  is one complete program implementing the direction, not an individual edit.

The agent constructs candidates from inspectable proposal data  $\mathcal{D}_{\text{prop}}$ , using the feedback table to connect evaluation errors to artifacts and execution traces. A disjoint sealed subset  $\mathcal{D}_{\text{dev}}$  returns aggregate measurements only. The direction is the unit of continued exploration; the complete program is the unit of selection. Connected edits share one parent and one measured outcome, so a bundle’s gain is not independent evidence for each edit.

## 5.3 CANDIDATE ACCEPTANCE AND PROGRAM INHERITANCE

Let  $B_t$  be the best program accepted so far and  $W_t$  the working program for the active direction. A new direction starts with  $W_t = B_t$ ; within that direction, the next candidate edits  $W_t$  using retained diagnostics. Every valid, distinct candidate receives the same development measurement protocol.

Define  $\Gamma_t(C)$  as the acceptance guard: complete and comparable execution, nondecreasing coverage, and the prespecified stratum-regression check. Writing  $\hat{J}_{\text{dev}}(C)$  for the recorded agreement of program  $C$ , acceptance is

$$a_t(C) = \mathbb{I}[\hat{J}_{\text{dev}}(C) > \hat{J}_{\text{dev}}(B_t)] \mathbb{I}[\Gamma_t(C)]. \quad (6)$$

A separate predicate  $w_t(C)$  determines whether the candidate can be inherited as the next working program:

$$B_{t+1} = \begin{cases} C, & a_t(C) = 1, \\ B_t, & \text{otherwise,} \end{cases} \quad W_{t+1} = \begin{cases} C, & w_t(C) = 1, \\ B_t, & \text{otherwise.} \end{cases} \quad (7)$$

Set  $w_t(C) = 0$  if required local or development comparisons are missing, incomplete, or incomparable. Otherwise,  $w_t(C) = 1$  for accepted candidates, candidates without observed damage, or damaged candidates with fresh, previously uncredited progress and preserved protection cases. Damage means a local regression or net regression against the parent or best program on sealed development comparisons. Progress is a newly repaired root error or a predeclared exact trace-field change; neither automatically verifies the hypothesis (Appendix D). Both decisions use the pre-update state, so failed inheritance returns to  $B_t$  even if  $C$  becomes  $B_{t+1}$ .

Equation 7 preserves nondecreasing recorded development agreement for  $B_t$  while allowing  $W_t$  to pass through candidates that do not improve it. This is a selection invariant, not a guarantee of unseen agreement or statistical significance.

## 5.4 CONTINUING A DIRECTION

Depth-first allocation gives the active direction the next editing opportunity rather than treating candidate rejection as a reason to switch directions. The agent uses diagnostic results to distinguish an incomplete implementation from a correction that did not help despite being implemented. These diagnostics guide further edits without guaranteeing eventual improvement.

Direction continuation does not require inheriting the latest candidate. When inheritance fails, the agent can keep working on the same direction from  $B_t$ , retaining the rejected candidate’s diagnostic history. It can request diagnostics, submit another implementation, or pause when no different informative test is proposed. The controller enforces program-inheritance eligibility and resource limits, not the agent’s interpretation of failure.

Judging a direction unhelpful requires diagnostic results that contradict its hypothesis within its stated scope, after relevant implementation problems have been examined. The controller does not make this judgment automatically, and resource exhaustion leaves the direction unresolved. Adaptive reuse makes  $\mathcal{D}_{\text{dev}}$  development data, not an independent test. On normal completion, Train orders eligible candidates; terminal  $\mathcal{D}_{\text{val}}$  assesses the frozen shortlist without reranking or edits. The seed remains if no eligible improvement exists, and independent Test/OOD assessment follows method freeze (Algorithm 1; Appendix C).

## 6 EXPERIMENTS

The experiments ask whether the evaluator improves judgments and downstream outputs (C1–C3), and whether preference mining and direction continuation account for those improvements under matched resources. Readiness determines inclusion from the 22-domain scope, with a separate evaluator per standard (Appendix A). Existing datasets supply candidate outputs for mining or reusable preferences; missing preferences are collected on existing or new outputs.

### 6.1 CORE RESULTS

**C1: evaluator alignment and selection.** Table 1 compares IterEval with metric references, static judges, and alternative adaptation methods. Controlled learned comparisons share feedback and the initial model (Appendices G and J). For each test input, all evaluators receive the same output group, with a default design of  $N = 4$ . Complete human rankings support global pairwise agreement, top-region agreement involving the human top-two outputs, Best-of- $N$  selection, and selection regret. When only ranks are available, normalized regret is  $(\text{rank}(\hat{y}) - 1)/(N - 1)$  for the selected output  $\hat{y}$ . A pair-only dataset supports pairwise evaluation, not an  $N$ -way selection claim.

Source-disjoint ID/OOD tests preserve the evaluation standard and sample independently of mining conflicts. For a fixed evaluator, paired effects use 95% intervals over original-input groups; independent searches assess optimizer variation separately. Appendix F specifies coverage, worst strata, repeated scoring, and uncertainty.

**C2: critic-guided refinement.** For each unseen input, a frozen generator produces one fresh output  $y_0$ . The seed and IterEval critics inspect that same output with matched evidence and critique formats. The same generator refines each once under identical prompts, decoding, tools, and token budgets. New blinded rankings compare IterEval-refined with Seed-refined and assess both against  $y_0$ , including major regressions. SV4 and IG1 are priorities, with TG1 testing continuity across tasks (Table 8).

**C3: reward-guided training.** Base, Seed-GRPO, and IterEval-GRPO share an initial checkpoint, with LoRA and group relative policy optimization for the trained arms (Hu et al., 2022; Shao et al., 2024). Rewards are frozen before training. Rollouts are sampled online by each arm’s own evolving policy and are not shared. The arms match training inputs, policy-update resources, and reward-call budgets. New blinded judgments and reserved external metrics assess their fresh outputs. C3 targets IG1, SV2, and the TG4 control after reward diagnostics (Appendix I). C3 remains a design-only study here; no parameter training or training benefit is reported.

### 6.2 ABLATION STUDIES

Search comparisons hold feedback fixed and count all proposal, diagnostic, measurement, and validation costs. The decisive contrast is direction continuation versus history-matched proposals that always edit the best program (Figure 5a). The primary matching variable is total model-and-tool cost under frozen prices, not the number of attempts. Sampling policies are compared at equal human budgets  $N_H \in \{20, 50, 100, 200\}$ , measured in original-input annotation units. The endpoint is final held-out alignment under the same evolution budget, with random anchors charged to every arm (Figure 5b). The decisive sampling contrast is preference mining versus random sampling with fixed feedback detail. Equal units test label efficiency; measured annotation time is needed to test human efficiency. Same-pair feedback-granularity tests and compatible component replacements complete the comparisons in Appendix J.



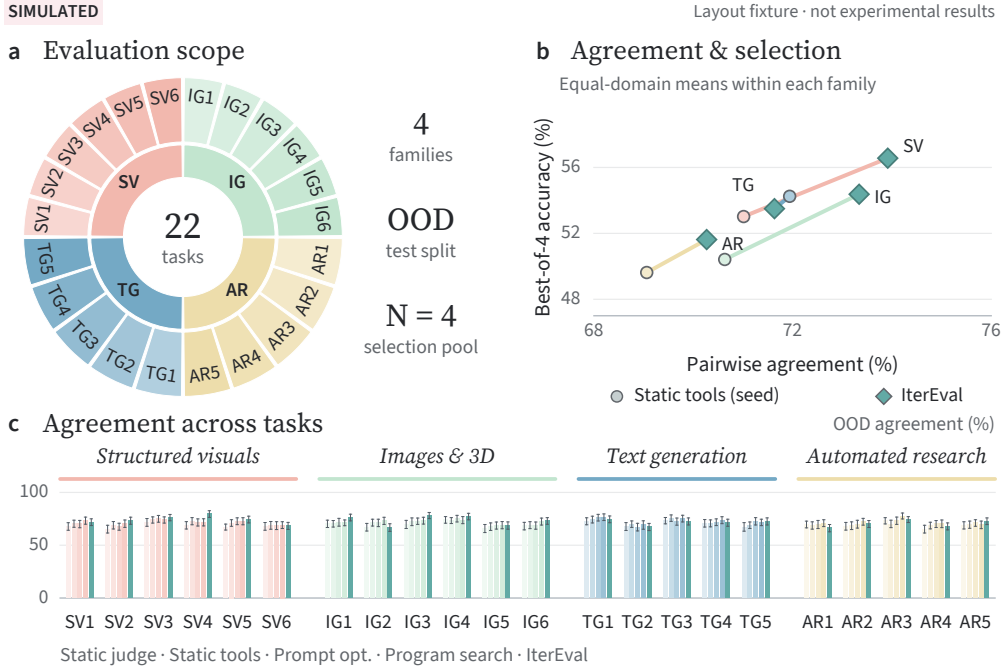


Figure 3: **C1 across tasks: simulated reporting layout, not experimental results.** Panel (a) maps 22 tasks into four families; equal family quadrants indicate taxonomy, not sample shares. Panel (b) links seed and IterEval family means for pairwise agreement and Best-of-4 selection, without implying a causal relation. Panel (c) reports five methods on each task using their jointly complete OOD inputs. Whiskers are 95% input-bootstrap intervals; selection counts failed scoring as failure. Table 1 specifies the full empirical comparison. Appendix L documents denominators and provenance.

### 6.3 ANALYSIS AND STUDIES

Trajectory analysis links failed implementations, diagnoses, subsequent edits, and outcomes, including exhausted directions. Paired interventions test missing observations versus their incorrect use, while same-subset comparisons assess feedback conversion. Appendix K separates these analyses from annotation, search, and deployment costs.

## 7 RESULTS AND ANALYSIS

**Draft status.** Table 1 is unmeasured; Figure 3 uses simulated records to illustrate task-level reporting. The interpretations below are conditional, not observed findings. Downstream and budget placeholders appear in Appendices H and J.

**C1: agreement that improves selection.** The favorable pattern would be higher ID/OOD agreement together with better Best-of- $N$  selection and lower regret (Table 1). Figure 3b–c separates selection from task-level agreement; an agreement gain without better selection would limit the downstream claim. Cross-domain gains would support adapting the procedure to different standards, not transferring one evaluator universally.

**C2: critique that improves the output.** More human-preferred refinements, accompanied by fewer major regressions, would indicate that the evolved evaluator provides more useful guidance

Table 1: **C1 main comparison: unmeasured placeholders.** Pair/Top: global/top-region agreement; BoN: Best-of- $N$  accuracy; Reg: normalized rank regret. Cells await equal-domain means and input-group 95% intervals for one fixed search replicate. Methods share cohorts per endpoint; eligible-domain counts must accompany results, with pair-only corpora entering Pair only. Dashes are pending, not zeros; shading is not a ranking. Appendix H specifies disaggregation and separate search variation.

| Evaluator                | ID test         |                |                |                  | OOD test        |                |                |                  |
|--------------------------|-----------------|----------------|----------------|------------------|-----------------|----------------|----------------|------------------|
|                          | Pair $\uparrow$ | Top $\uparrow$ | BoN $\uparrow$ | Reg $\downarrow$ | Pair $\uparrow$ | Top $\uparrow$ | BoN $\uparrow$ | Reg $\downarrow$ |
| Best single metric       | —               | —              | —              | —                | —               | —              | —              | —                |
| Metric ensemble          | —               | —              | —              | —                | —               | —              | —              | —                |
| Static model judge       | —               | —              | —              | —                | —               | —              | —              | —                |
| Static tool judge (seed) | —               | —              | —              | —                | —               | —              | —              | —                |
| Score calibration        | —               | —              | —              | —                | —               | —              | —              | —                |
| Evaluator fine-tuning    | —               | —              | —              | —                | —               | —              | —              | —                |
| Prompt optimization      | —               | —              | —              | —                | —               | —              | —              | —                |
| Program search           | —               | —              | —              | —                | —               | —              | —              | —                |
| IterEval                 | —               | —              | —              | —                | —               | —              | —              | —                |

(Table 8). Changes relative to  $y_0$  would distinguish genuine improvement from a less damaging revision.

**C3: reward that improves generation.** A favorable outcome would combine human preference for IterEval-GRPO over Seed-GRPO with retained factuality, coverage, and validity. Higher reward alone would be insufficient; comparison with Base must also establish whether training improves generation. Conclusions would remain task- and policy-specific, with external checks reported separately (Table 9).

**Continuation rather than extra search.** Higher held-out agreement than restarting and history-only proposals at matched total cost would support depth-first allocation (Figure 5a). To support the proposed mechanism, trajectories must also connect a failed implementation, a diagnosed obstacle, and a later correction within the same direction.

**Informative preferences rather than more labels.** A favorable sampling result would reach the same held-out agreement with fewer annotation units, or higher agreement at the same units (Figure 5b). Equal-label gains that disappear at equal annotation time would not establish greater human efficiency.

## 8 DISCUSSION

Depth-first evaluator evolution permits continuation without lowering the recorded best development score, but continuation has an opportunity cost. If history-only search beats independent proposals yet matches our search at equal cost, retained diagnostics may explain gains without direction continuation. Development-only gains would instead suggest adaptation to repeatedly inspected feedback, not better evaluation of unseen outputs. Neither outcome supports a general advantage for depth-first allocation.

Downstream use changes which errors matter. Selection depends on ordering the best outputs; critique requires actionable diagnoses. Training shifts outputs toward whatever the reward favors, so C1 agreement or C2 refinement cannot substitute for C3 evaluation.

### 8.1 LIMITATIONS AND FUTURE WORK

First, consensus can hide shared errors, requiring random checks and measured human-time comparisons. Second, diagnostics do not verify a hypothesis; independent trajectory review must include

failed and exhausted directions. Finally, sampling efficiency and continuation benefits remain unmeasured, requiring matched-budget runs before empirical claims.

## 9 CONCLUSION

Human preferences must be turned into evaluation procedures that work on new outputs. IterEval combines disagreement-based preference mining to reduce redundant annotation with depth-first evaluator evolution that separates candidate acceptance from direction continuation. Matched-budget comparisons test whether these choices improve evaluation, while separate downstream tests assess selection, refinement, and reward-guided training. The present placeholders and simulated illustrations provide no empirical evidence of these benefits.

## AI USE STATEMENT

Generative AI assisted manuscript drafting, literature organization, and the preparation of editable LaTeX conceptual diagrams and plotting code. Figure 1 uses AI-generated illustrative images from an earlier figure-design pass, with editable vector labels and search paths. Its preferences and intermediate outcomes are constructed, not observed annotations or optimization traces. The photograph in Figure 2a is also AI-generated and illustrative. The quantitative displays use explicitly simulated data, generated offline for reviewing the reporting design. They are not observed cases, human annotations, model executions, or performance measurements. No empirical results or parameter-training runs were produced for this draft. Bibliographic identities were checked against primary metadata sources. Author review, responsibility for the final content, and the final disclosure of research-stage AI use remain to be completed before submission.

## ETHICS STATEMENT

Human preferences can encode annotator and model biases rather than a universal definition of quality. Domain-specific criteria and uncertainty should remain visible, especially for automated research judgments. Any new annotation study must document consent, compensation, privacy protection, and the applicable institutional review determination before data collection. Artifact licenses and source permissions must be checked independently of technical accessibility. Generated code and evidence tools require restricted execution and controlled external access. Reward-guided optimization requires checks for exploitation of the evaluator and degradation outside its favored criteria. No private artifacts, identifiable annotation records, or experimental traces are included in this working manuscript. Study-specific approvals and release decisions are pending.

## REPRODUCIBILITY STATEMENT

Sections 3–6 define semantic mappings, feedback, the evaluation program, evolution, and controlled evaluation. Appendices B–F specify annotation, access boundaries, the algorithm, and measurement. Appendices G–M specify adaptation controls, core reporting, planned training, ablations, analyses, simulated fixtures, and the experiment manifest. Appendix L identifies the offline generator and provenance for the numerical displays. The current manuscript contains no empirical results to reproduce. Before submission, the authors must supply a versioned, appropriately anonymized implementation and permitted data-processing and analysis artifacts with completed manifests. Their availability is not asserted here.

## REFERENCES

- Lakshya A Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziemis, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alex Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning. In *International Conference on Learning Representations*, volume 2026, pp. 8479–8565, 2026. URL [https://proceedings.iclr.cc/paper\\_files/paper/2026/hash/0e9e708b6f48e14fd0ac29e167413f76-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2026/hash/0e9e708b6f48e14fd0ac29e167413f76-Abstract-Conference.html).
- Boshui Chen, Huiping Liu, and Shaolei Zhang. WebGrader: Training LLMs for web development with self-evolving programmatic grader. *arXiv preprint arXiv:2608.06474*, 2026a. URL <https://arxiv.org/abs/2608.06474>.
- Xin-Sheng Chen, Jiayu Zhu, Pei lin Li, Hanzheng Wang, Shuojin Yang, and Meng-Hao Guo. PresentBench: A Fine-Grained Rubric-Based Benchmark for Slide Generation. *arXiv preprint arXiv:2603.07244*, 2026b. URL <https://arxiv.org/abs/2603.07244>.
- Alexander R. Fabbri, Wojciech Kryściński, Bryan McCann, Caiming Xiong, Richard Socher, and Dragomir Radev. SummEval: Re-evaluating Summarization Evaluation. *Transactions of the Association for Computational Linguistics*, 9:391–409, 2021. doi: 10.1162/tacl\_a\_00373.

- Yuru Feng, Yaoqi Chen, Beidi Zhao, Qianxi Zhang, Xinjiang Wang, Jianan Lu, Zhirui Wang, Shusen Xu, Zengzhong Li, and Qi Chen. SkillHEX: Improving agent skills via hypothesis-driven autonomous exploration and exploitation. *arXiv preprint arXiv:2608.05628*, 2026. URL <https://arxiv.org/abs/2608.05628>.
- Dhruba Ghosh, Hannaneh Hajishirzi, and Ludwig Schmidt. GenEval: An object-focused framework for evaluating text-to-image alignment. In *Advances in Neural Information Processing Systems*, volume 36, pp. 52132–52152, 2023. doi: 10.52202/075280-2270.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=nZeVKeeFYf9>.
- Shengran Hu, Cong Lu, and Jeff Clune. Automated Design of Agentic Systems. In *International Conference on Learning Representations*, volume 2025, pp. 21344–21377, 2025. URL [https://proceedings.iclr.cc/paper\\_files/paper/2025/hash/36b7acf6f6010652b3f2a433774a66fe-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2025/hash/36b7acf6f6010652b3f2a433774a66fe-Abstract-Conference.html).
- Xinyu Hu, Mingqi Gao, Sen Hu, Yang Zhang, Yicheng Chen, Teng Xu, and Xiaojun Wan. Are LLM-based evaluators confusing NLG quality criteria? In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 9530–9570. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.516. URL <https://aclanthology.org/2024.acl-long.516/>.
- Zisu Huang, Jingwen Xu, Yifan Yang, Ziyang Gong, Qihao Yang, Muzhao Tian, Xiaohua Wang, Changze Lv, Xuemei Gao, Qi Dai, Bei Liu, Kai Qiu, Xue Yang, Dongdong Chen, Xiaoqing Zheng, and Chong Luo. From Raw Experience to Skill Consumption: A Systematic Study of Model-Generated Agent Skills. *arXiv preprint arXiv:2605.23899*, 2026. URL <https://arxiv.org/abs/2605.23899>.
- Pei Ke, Bosi Wen, Andrew Feng, Xiao Liu, Xuanyu Lei, Jiale Cheng, Shengyuan Wang, Aohan Zeng, Yuxiao Dong, Hongning Wang, Jie Tang, and Minlie Huang. CritiqueLLM: Towards an informative critique generation model for evaluation of large language model generation. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 13034–13054. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.704. URL <https://aclanthology.org/2024.acl-long.704/>.
- Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T. Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. DSPy: Compiling Declarative Language Model Calls into Self-Improving Pipelines. In *International Conference on Learning Representations*, volume 2024, pp. 54928–54958, 2024. URL [https://proceedings.iclr.cc/paper\\_files/paper/2024/file/f1cf02ce09757f57c3b93c0db83181e0-Paper-Conference.pdf](https://proceedings.iclr.cc/paper_files/paper/2024/file/f1cf02ce09757f57c3b93c0db83181e0-Paper-Conference.pdf).
- Seungone Kim, Jamin Shin, Yejin Cho, Joel Jang, Shayne Longpre, Hwaran Lee, Sangdoo Yun, Seongjin Shin, Sungdong Kim, James Thorne, and Minjoon Seo. Prometheus: Inducing Fine-grained Evaluation Capability in Language Models. In *International Conference on Learning Representations*, volume 2024, pp. 29927–29962, 2024a. URL [https://proceedings.iclr.cc/paper\\_files/paper/2024/file/803485352e61e3ebf41221e4776c9fd4-Paper-Conference.pdf](https://proceedings.iclr.cc/paper_files/paper/2024/file/803485352e61e3ebf41221e4776c9fd4-Paper-Conference.pdf).
- Tae Soo Kim, Yoonjoo Lee, Jamin Shin, Young-Ho Kim, and Juho Kim. EvalLM: Interactive evaluation of large language model prompts on user-defined criteria. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, pp. 1–21. ACM, 2024b. doi: 10.1145/3613904.3642216. URL <https://doi.org/10.1145/3613904.3642216>.
- Yuval Kirstain, Adam Polyak, Uriel Singer, Shahbuland Matiana, Joe Penna, and Omer Levy. Pick-a-Pic: An Open Dataset of User Preferences for Text-to-Image Generation. In *Advances in Neural Information Processing Systems*, volume 36, pp. 36652–36663, 2023. doi: 10.52202/075280-1594.

- Yuxuan Liu, Tianchi Yang, Shaohan Huang, Zihan Zhang, Haizhen Huang, Furu Wei, Weiwei Deng, Feng Sun, and Qi Zhang. Calibrating LLM-based evaluator. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pp. 2638–2656. ELRA and ICCL, 2024. doi: 10.63317/3posqwhnb6g. URL <https://aclanthology.org/2024.lrec-main.237/>.
- Xiaotian Luo, Dizhan Xue, Fengxingyu Wang, Chuanrui Hu, and Yafeng Deng. HarnessBank: Semantic gene-bank search with gated verification for agent-harness self-evolution. *arXiv preprint arXiv:2607.13683*, 2026. URL <https://arxiv.org/abs/2607.13683v2>.
- Changze Lv, Jie Zhou, Wentao Zhao, Jingwen Xu, Shihan Dou, Zisu Huang, Muzhao Tian, Xiaohua Wang, Zhenggang Guo, Yang Liu, Pluto Zhou, Tao Gui, Le Tian, Xiao Zhou, Xiaoqing Zheng, Xuanjing Huang, and Jie Zhou. Learning Query-Specific Rubrics from Human Preferences for DeepResearch Report Generation. *arXiv preprint arXiv:2602.03619*, 2026. URL <https://arxiv.org/abs/2602.03619>.
- William Muldrew, Peter Hayes, Mingtian Zhang, and David Barber. Active preference learning for large language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 36577–36590. PMLR, 2024. URL <https://proceedings.mlr.press/v235/muldrew24a.html>.
- Minhyeon Oh, Seungjoon Lee, and Jungseul Ok. Comparison-based active preference learning for multi-dimensional personalization. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 33145–33166. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-long.1590. URL <https://aclanthology.org/2025.acl-long.1590/>.
- Qi Qian, Chengsong Huang, Jingwen Xu, Changze Lv, Muling Wu, Wenhao Liu, Xiaohua Wang, Zhenghua Wang, Zisu Huang, Muzhao Tian, Jianhan Xu, Kun Hu, He-Da Wang, Yao Hu, Xuanjing Huang, and Xiaoqing Zheng. Benchmark<sup>2</sup>: Systematic Evaluation of LLM Benchmarks. *arXiv preprint arXiv:2601.03986*, 2026. URL <https://arxiv.org/abs/2601.03986>.
- H. S. Seung, M. Oppen, and H. Sompolinsky. Query by committee. In *Proceedings of the Fifth Annual Workshop on Computational Learning Theory*, pp. 287–294. ACM, 1992. doi: 10.1145/130385.130417. URL <https://doi.org/10.1145/130385.130417>.
- Shreya Shankar, Haotian Li, Parth Asawa, Madelon Hulsebos, Yiming Lin, J. D. Zamfirescu-Pereira, Harrison Chase, Will Fu-Hinthorn, Aditya G. Parameswaran, and Eugene Wu. spade: Synthesizing data quality assertions for large language model pipelines. *Proceedings of the VLDB Endowment*, 17(12):4173–4186, 2024a. doi: 10.14778/3685800.3685835. URL <https://www.vldb.org/pvldb/vol17/p4173-shankar.pdf>.
- Shreya Shankar, J.D. Zamfirescu-Pereira, Bjoern Hartmann, Aditya Parameswaran, and Ian Arawjo. Who Validates the Validators? Aligning LLM-Assisted Evaluation of LLM Outputs with Human Preferences. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–14, 2024b. doi: 10.1145/3654777.3676450.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024. URL <https://arxiv.org/abs/2402.03300>.
- Chenglei Si, Yanzhe Zhang, Ryan Li, Zhengyuan Yang, Ruibo Liu, and Diyi Yang. Design2Code: Benchmarking Multimodal Code Generation for Automated Front-End Engineering. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 3956–3974, 2025. doi: 10.18653/v1/2025.naacl-long.199.
- Xiaohua Wang, Muzhao Tian, Yuqi Zeng, Zisu Huang, Jiakang Yuan, Bowen Chen, Jingwen Xu, Mingbo Zhou, Wenhao Liu, Muling Wu, Zhenggang Guo, Qi Qian, Yifei Wang, Feiran Zhang, Ruicheng Yin, Shihan Dou, Changze Lv, Tao Chen, Kaitao Song, Xu Tan, Tao Gui,

- Xiaoqing Zheng, and Xuanjing Huang. Reward Hacking in the Era of Large Models: Mechanisms, Emergent Misalignment, Challenges. *arXiv preprint arXiv:2604.13602*, 2026. URL <https://arxiv.org/abs/2604.13602>.
- Jiazheng Xu, Xiao Liu, Yuchen Wu, Yuxuan Tong, Qinkai Li, Ming Ding, Jie Tang, and Yuxiao Dong. ImageReward: Learning and Evaluating Human Preferences for Text-to-Image Generation. In *Advances in Neural Information Processing Systems*, volume 36, pp. 15903–15935, 2023. doi: 10.52202/075280-0700.
- Yifan Yang, Ziyang Gong, Weiquan Huang, Qihao Yang, Ziwei Zhou, Zisu Huang, Yan Li, Xuemei Gao, Qi Dai, Bei Liu, Kai Qiu, Yuqing Yang, Dongdong Chen, Xue Yang, and Chong Luo. SkillOpt: Executive strategy for self-evolving agent skills. *arXiv preprint arXiv:2605.23904*, 2026. URL <https://arxiv.org/abs/2605.23904v2>.
- Seonghyeon Ye, Doyoung Kim, Sungdong Kim, Hyeonbin Hwang, Seungone Kim, Yongrae Jo, James Thorne, Juho Kim, and Minjoon Seo. FLASK: Fine-grained language model evaluation based on alignment skill sets. In *International Conference on Learning Representations*, volume 2024, pp. 55361–55414, 2024. URL [https://proceedings.iclr.cc/paper\\_files/paper/2024/hash/f41b4a6b202adcd8e150a9d4f124d8f6-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2024/hash/f41b4a6b202adcd8e150a9d4f124d8f6-Abstract-Conference.html).
- Mert Yuksekgonul, Federico Bianchi, Joseph Boen, Sheng Liu, Pan Lu, Zhi Huang, Carlos Guestrin, and James Zou. Optimizing generative AI by backpropagating language model feedback. *Nature*, 639(8055):609–616, 2025. doi: 10.1038/s41586-025-08661-4.
- Jenny Zhang, Shengran Hu, Cong Lu, Robert Lange, and Jeff Clune. Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents. In *International Conference on Learning Representations*, volume 2026, pp. 104223–104294, 2026. URL [https://proceedings.iclr.cc/paper\\_files/paper/2026/hash/aa5f5e6eb6f613ec412f1d948dfa21a5-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2026/hash/aa5f5e6eb6f613ec412f1d948dfa21a5-Abstract-Conference.html).
- Hao Zheng, Xinyan Guan, Hao Kong, Wenkai Zhang, Jia Zheng, Weixiang Zhou, Hongyu Lin, Yaojie Lu, Xianpei Han, and Le Sun. PPTAgent: Generating and Evaluating Presentations Beyond Text-to-Slides. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 14402–14418, 2025. doi: 10.18653/v1/2025.emnlp-main.728.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, Hao Zhang, Joseph Gonzalez, and Ion Stoica. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, volume 36, pp. 46595–46623, 2023. doi: 10.52202/075280-2020.
- Mingchen Zhuge, Changsheng Zhao, Dylan R. Ashley, Wenyi Wang, Dmitrii Khizbullin, Yungang Xiong, Zechun Liu, Ernie Chang, Raghuraman Krishnamoorthi, Yuandong Tian, Yangyang Shi, Vikas Chandra, and Jürgen Schmidhuber. Agent-as-a-judge: Evaluate agents with agents. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 80569–80611. PMLR, 2025. URL <https://proceedings.mlr.press/v267/zhuge25a.html>.

## A DOMAIN PANORAMA AND EVALUATION STANDARDS

The research scope is organized by evaluation standard, not by file format. Table 2 retains all 22 candidate domains. Its dimensions are a starting taxonomy to be refined from domain evaluation knowledge before selecting datasets. Each instantiated domain must additionally fix priorities, decision rules, and an annotation protocol. Inclusion in this panorama does not establish data readiness, an implemented adapter, or empirical transfer. Figure 2a gives representative artifacts; the table makes their differing evaluation criteria explicit.

Table 2: Complete domain panorama. The four families organize the scope; every row requires its own standard and readiness audit. These are study candidates, not 22 completed evaluations.

| ID                                        | Domain                                               | Core quality considerations                                                                      |
|-------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Structured visual artifacts</b>        |                                                      |                                                                                                  |
| TD-SV1                                    | Brief-conditioned single-slide generation            | Brief coverage, hierarchy, readability, composition, and visual consistency.                     |
| TD-SV2                                    | Screenshot-conditioned webpage reconstruction        | Reference content, spatial arrangement, typography, and visual fidelity.                         |
| TD-SV3                                    | Data-to-visualization generation                     | Data fidelity, valid encodings, readability, and communicative usefulness.                       |
| TD-SV4                                    | Specification-conditioned webpage generation         | Requirement satisfaction, functional behavior, usability, and visual organization.               |
| TD-SV5                                    | Source-grounded slide-deck generation                | Source faithfulness, narrative, coverage, slide readability, and deck consistency.               |
| TD-SV6                                    | Topic-conditioned slide-deck generation              | Relevance, factual support, narrative completeness, audience fit, and presentation quality.      |
| <b>Image generation and understanding</b> |                                                      |                                                                                                  |
| TD-IG1                                    | Detailed image-caption generation                    | Grounding, coverage, relation accuracy, specificity, and absence of hallucination.               |
| TD-IG2                                    | Text-to-image generation                             | Prompt alignment, object and relation accuracy, visual coherence, and preference.                |
| TD-IG3                                    | Single-source instruction-based image editing        | Instruction fulfillment, preservation of unrequested content, and image quality.                 |
| TD-IG4                                    | LR-conditioned image super-resolution                | Reference fidelity when available, perceptual detail, artifact avoidance, and plausible texture. |
| TD-IG5                                    | Text-to-3D object generation                         | Prompt correspondence, geometry, appearance, and multi-view consistency.                         |
| TD-IG6                                    | Image-to-3D object generation                        | Source-view fidelity, plausible unseen geometry, appearance, and multi-view consistency.         |
| <b>Text generation</b>                    |                                                      |                                                                                                  |
| TD-TG1                                    | Source-grounded document summarization               | Faithfulness, relevance, coverage, coherence, and concision.                                     |
| TD-TG2                                    | RAG-supported long-form answer generation            | Answer correctness, evidence grounding, attribution, completeness, and organization.             |
| TD-TG3                                    | Machine translation                                  | Meaning preservation, omissions and additions, fluency, terminology, and style.                  |
| TD-TG4                                    | Instruction-constraint-following response generation | Constraint satisfaction, substantive correctness, relevance, and response quality.               |
| TD-TG5                                    | Open-ended dialogue-response generation              | Contextual appropriateness, helpfulness, consistency, safety, and conversational quality.        |
| <b>Automated research</b>                 |                                                      |                                                                                                  |
| TD-AR1                                    | Research-idea generation                             | Novelty relative to evidence, significance, feasibility, and testability.                        |
| TD-AR2                                    | ML/NLP ablation-plan generation                      | Hypothesis identification, controlled contrasts, confound management, and feasibility.           |
| TD-AR3                                    | Table-to-insight report generation                   | Numerical faithfulness, appropriate inference, uncertainty, and relevance.                       |



Table 2 continued

| ID     | Domain                          | Core quality considerations                                                                |
|--------|---------------------------------|--------------------------------------------------------------------------------------------|
| TD-AR4 | Related-work generation         | Coverage, citation correctness, faithful comparison, and synthesis.                        |
| TD-AR5 | Peer-review critique generation | Technical accuracy, evidence, significance assessment, fairness, and actionable criticism. |

Representative resources include PPTAgent and PresentBench for presentations, Design2Code for screenshot-conditioned webpages, SummEval for summarization, and image-preference and alignment resources (Zheng et al., 2025; Chen et al., 2026b; Si et al., 2025; Fabbri et al., 2021; Kirstain et al., 2023; Xu et al., 2023; Ghosh et al., 2023). Their outputs, references, metrics, and annotations have different evidential roles; a resource is not automatically a source of the required human preference. A benchmark collection spanning several standards remains a dataset collection, not an additional domain. Research execution with a complete executable postcondition is excluded, while ideation, study design, interpretation, writing, and critique remain open-ended evaluation problems.

**Dataset roles.** Dev supports data mining and evaluator development, ID tests unseen inputs from the development source distribution, and OOD changes source distribution while preserving the standard. External datasets provide separately locked assessment only and do not determine semantic mappings or data mining. Harder examples from the same source are a difficulty stratum, not a new OOD domain. A domain selected for confirmatory execution must pass the audit below; selection does not remove unselected domains from the research scope.

## B FEEDBACK PROVENANCE, ANNOTATION, AND READINESS

The two construction settings in Table 3 share the same evaluator-evolution problem. They differ in the work needed to obtain matched human feedback. A domain may use both settings. Direct, derived, and materials-only provenance describe what a resource supplies; they are not additional domain classes or alternative algorithms.

Inputs, references, editable artifacts, and automatic metrics without matched human judgments are materials only. They may support output construction or independent assessment, but do not supply human feedback. When  $H_0$  is absent, we preserve the original task and evaluation standard, separate source groups, recover readable outputs or generate them with frozen generators, and collect the required feedback. Preparation can precede annotation; final human-alignment claims cannot precede matched human assessment.

**Annotation units and levels.** One annotation unit is an original input and its anonymized output group, including permitted references. Pairs derived from that group remain dependent. The core tuple in Equation 1 is retained together with dataset and license, source grouping, generator and prompt version, collection time, annotator pool, agreement, and conversion provenance. Generator identity is recorded for audit but hidden from annotators and the deployed scorer.

The planned evolution protocol collects  $H_0$  and core  $H_1$ , with  $H_2$  on conflict and failure subsets.  $H_3$  is optional. Two independent annotators with adjudication may support development; final ID/OOD assessment targets three independent annotators and blinded complete rankings. Core dimensions and the tie policy are fixed in advance. A model or an author decision is not substituted for a missing independent final annotation. An  $H_0$ -only setting must be reported as such; the framework’s richer interface does not establish that a run actually received  $H_1$  or  $H_2$ .

**Agreement, abstention, and reannotation.** New labels are not assumed to be more reliable than reused labels. Report the agreement statistic, its valid-response denominator, and the number of raters per group. Abstention uses all requested ratings as its denominator, including those excluded from valid-pair agreement. Group outcomes distinguish decisive preference, tie, no consensus, and insufficient votes. The simulated audit in Figure 9a–c illustrates these separate quantities. Individual disagreements are retained rather than filtered to match an old winner.

Table 3: Two routes to human feedback. Report each source’s protocol, agreement, abstention, and available human feedback separately. An adapter changes representation, not the quality construct or the provenance of a judgment.

| Setting            | Label origin and protocol                                                                                                                                  | Human feedback                                                                                                                                       |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reused preferences | Original human pairs/rankings on readable same-input outputs; retain the source protocol and document any conversion from rankings, scores, or errors      | $H_0$ only for overall judgments; $H_1/H_2$ only when actually annotated. Unknown source agreement or abstention remains unknown                     |
| New preferences    | Freeze the task, visible evidence, dimensions, ties, and abstention before collecting independent blind judgments on existing or freshly generated outputs | Collect the levels required by the study; preserve individual votes and unresolved disagreements. Fresh C2/C3 outputs require new endpoint judgments |

Annotators do not see model scores, method identity, or earlier winners. A changed rubric or visible evidence defines a changed protocol; resulting differences do not by themselves establish errors in the older labels.

**Label-blind preference mining.** Freeze domain knowledge, signal definitions, semantic mappings, and the calibration recipe before accessing preference labels. Select a random anchor, reveal only its authorized training judgments to estimate reliability, then freeze the targeted sampling policy before opening labels in the remaining pool. Charge anchor annotations to every strategy’s human budget. An existing labeled dataset must simulate the same access order. The final cohort is sampled independently of conflicts and never contributes to reliability calibration.

**Implemented conflict ranking.** Preference mining computes structural disagreement without reading the pair’s human preference. Let  $I_{\text{within},q}(r)$  indicate opposing signals from distinct source groups on dimension  $q$ ,  $I_{\text{overall},q}(r)$  indicate opposition between an overall and a dimensional signal, and  $I_{\text{tradeoff}}(r)$  indicate opposed dimension-level preferences. Equation 3 ranks these conflicts, assigning twice the weight to within-dimension disagreement. Pairs are ranked by decreasing  $P(r)$ , with stable pair identifiers breaking ties. The coefficients define a structural ranking, not estimated signal reliability or a probability of evaluation error. Overall–dimension comparisons retain whether their signals share a source group; they are not automatically independent evidence. Each source group contributes at most one preference sign to a dimension’s aggregate preference, with opposing within-group signs yielding abstention. Missing outputs and small-margin ties remain distinct from opposing preferences.

For repeated model signals, the pairwise signal margin is the median of same-repeat score differences between the two artifacts; variation and sign changes are retained. This differs from Equation 5, which first takes each artifact’s median final score and then compares them. A missing required signal repeat makes that evaluation signal unavailable for mining. Deterministic metrics use declared ranges and orientations, not ranges fitted to the output pool or human labels. The random anchor is selected by original-input hashes before scoring, preserving output groups. Label-blind execution does not establish label-blind upstream cohort selection; historical data must also document how the output pool was chosen.

**Human qualification and implementation boundary.** The implemented miner supplies the structural queue and its source-dependence records. The full preference-mining protocol additionally requires human-annotated anchors to qualify signals before targeted annotation; cross-domain human-reliability calibration and the equal-budget advantage remain unverified. Model-annotated pilot anchors measure agreement with that model, not human reliability. The annotation fraction, qualification thresholds, and targeted policy must be frozen for the human study, rather than inherited silently from a pilot. The feedback table  $Z$  is a logical join over feedback, artifact, signal, and execution records; absent human dimensions or rationales remain absent.

**Source and tool status.** Registry entries distinguish executable, planned, and blocked sources (Table 4). A publication or code link without compatible inputs, weights, or dependencies is not an executed signal. Available tools begin as candidates; a route must explicitly activate them. Source

Table 4: Readiness criteria before a domain enters confirmatory evaluation. These are audit requirements, not completed coverage scores.

| Audit object         | Required record and exclusion condition                                                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evaluation standard  | Open-endedness, dimensions and hierarchy, priorities, decision rule, and human protocol. Exclude tasks fully decided by an executable postcondition.                       |
| Knowledge provenance | Benchmarks, expert guidance, rubrics, and methods with versions, dates, input assumptions, known biases, and availability.                                                 |
| Source coverage      | Dimension $\times$ source-type matrix: deterministic, learned metric, LLM/VLM judge, external analysis tool. Each core dimension needs executable evidence.                |
| Source independence  | For conflict mining, ordinarily two mechanisms per dimension. Record shared models, training labels, metrics, and mixed signals; multiple names are not independent votes. |
| Executable artifacts | Recoverable input/output groups, readable artifacts, references, permissions, and source fingerprints. Missing assets do not become scorer disagreement.                   |
| Conflict pool        | Three semantic strata, availability, margins, repeat variance, anchor reliability, dependence caps, and uncovered strata. Noise alone is not a gap.                        |
| Feedback fidelity    | Direct versus derived provenance, agreement, mapping to $H_0/H_1/H_2$ , adapter checks on a common subset, and additional annotation needs.                                |
| Source isolation     | Original-input deduplication, Dev/ID/OOD/External roles, contamination review, and a pre-search External Metric Reserve.                                                   |

Table 5: Information available to evolution and assessment. Every invocation of the deployed scorer remains pointwise and label-blind, including on Train.

| Partition      | Information released                       | Permitted decision                                                    |
|----------------|--------------------------------------------|-----------------------------------------------------------------------|
| Proposal Train | Authorized artifacts, feedback, and traces | Mechanism hypotheses and candidate evolution                          |
| Sealed Train   | Prespecified aggregates only               | Candidate acceptance, program inheritance, and direction continuation |
| Terminal Val   | Frozen-shortlist measurements              | Terminal assessment; no reranking or further edits                    |
| Test/OOD       | Locked until method and analysis freeze    | Independent final reporting                                           |
| Metric reserve | No access during evolution                 | Final assessment only; no tools, critique, reward, or selection       |

independence is assessed at the signal level, including model, training-label, and metric dependencies. Model-only comparisons hold rubric, evidence, context, and reasoning budget fixed. Replacing both the scoring model and its evidence-collection agent instead compares full systems. Deterministic or fixed learned metrics are not duplicated for every judge model.

## C DATA ACCESS AND INDEPENDENT ASSESSMENT

Table 5 separates proposal evidence, adaptive development measurements, and terminal assessment. The scorer always receives the same pointwise input contract, regardless of the cohort in which it is invoked. Only the authorized diagnostic process sees human labels and pair identities.

**Training and validation.** Proposal-Train evidence may inform evolution directions, candidates, and allocation. Sealed Train reveals prespecified aggregate agreement, coverage, and stratum guards without examples, labels, or per-case errors. Aggregates remain adaptive development evidence, not a final performance estimate. The terminal shortlist and its order are fixed from training evidence before Val access, and every shortlisted outcome is reported. Repeated Val calls can assess stability but do not create an independent test cohort. Interrupted searches do not obtain a new terminal assessment merely because they stopped.

**External metrics.** The External Metric Reserve is frozen before search. Reserved metrics and their equivalent wrappers are excluded from the executable tool library, router, diagnostician, critique, reward, and checkpoint-selection criteria. If an endpoint was used for optimization, it loses

independent-assessment status and must be replaced or explicitly reported as an optimization metric. This restriction does not forbid using other existing metrics as tools.

## D DEPTH-FIRST EVALUATOR EVOLUTION AND ATTRIBUTION CONTRACT

Figure 4 separates the editable observation and judgment mechanisms. Algorithm 1 specifies their evolution and selection.

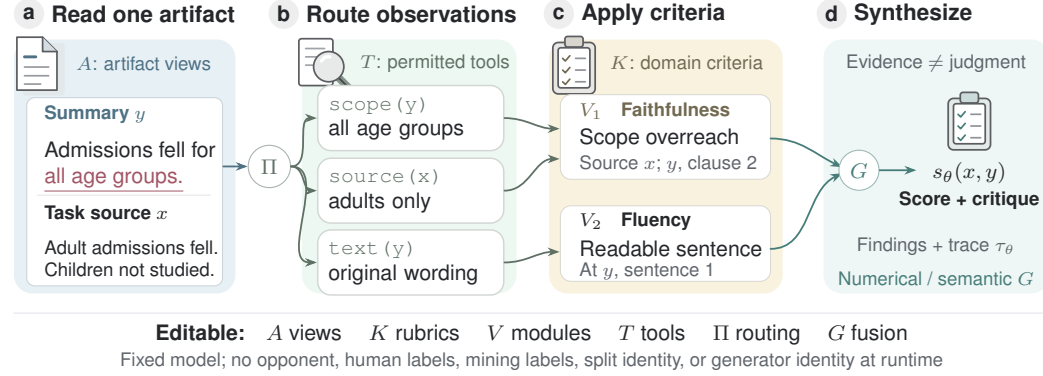


Figure 4: The six-component evaluation program. (a) Read one output and permitted task evidence. (b) Route tool observations. (c) Apply dimensional criteria. (d) Fuse sourced findings into a score and critique. The lower rail identifies the editable components. Human feedback guides evaluator evolution, never the deployed scorer. Artifact evidence and judgment remain separate; the summary and findings are a constructed example.

### Algorithm 1 Depth-first evaluator evolution: separate candidate acceptance from direction continuation

**Require:** Seed  $E_0$ , proposal feedback table  $Z_{\text{prop}}$ , sealed  $\mathcal{D}_{\text{dev}}$ , fixed search and execution limits

```

1: Measure  $E_0$ ; initialize best program  $B \leftarrow E_0$ , working program  $W \leftarrow E_0$ , and history
2: while search resources remain do
3:   Start a direction from  $B$ , or continue the active direction from  $W$ 
4:   Inspect proposal-Train artifacts and history; request diagnostics as needed
5:   Construct complete candidate  $C$  with its intent and inspected Train references
6:   if  $C$  is valid and distinct then
7:     Measure  $C$  on  $\mathcal{D}_{\text{dev}}$ ; retain local diagnostic outcomes
8:     Save pre-update best  $B_{\text{old}} \leftarrow B$ ; assess acceptance and inheritance against this state
9:     if the candidate-acceptance conditions hold then
10:       $B \leftarrow C$ 
11:     end if
12:      $W \leftarrow C$  if measured inheritance is eligible; otherwise  $W \leftarrow B_{\text{old}}$ 
13:   end if
14:   Retain changes, outcomes, failed attempts, and resource use in history
15:   Give the active direction the next opportunity to request diagnostics or submit an edit
16:   Pause on the agent's explicit request or an enforced resource/execution limit
17: end while
18: On normal completion, freeze and order eligible candidates using Train only
19: Assess a nonempty shortlist on terminal Val without reranking or further edits
20: Retain the Train-selected evaluator, or the seed if no eligible improvement exists

```

**Investigation and candidate construction.** The evolution agent reads original proposal-Train artifacts, installed component instructions, execution traces, and relevant history through a tool-driven editing session. It can request additional authorized proposal-Train measurements or fixed-input component probes. The submission contains actual edits, a short intent, and references to inspected training cases. Predicted orderings, contrast cases, and intermediate checks are optional declarations, not controller-inferred scientific predictions. The controller records the exact parent, installed

program, and measured outcomes. Local diagnostic success is not an admission requirement for the common development measurement.

An elementary edit changes one component; a connected candidate can assemble a tool, its route, and its consumer against a fixed parent. Only the complete candidate is executed and measured, not its partially assembled versions. Changed tools must be connected to valid routes, although valid routing alone does not establish correct use of their outputs. Interface validity, fixed-model constraints, and execution limits remain mandatory; edit-character counts are recorded as guidance rather than a hard acceptance or stopping rule. Unchanged programs and verified historical replays are distinguished from fresh trials, and failed calls remain in the cost record.

**Diagnostic results.** Fixed-input probes compare supported specialist or synthesis variants while holding their upstream inputs fixed. Such probes bind their observations to the exact tested candidate and inputs; changing a tool, route, or topology requires the corresponding complete execution. An optional exact intermediate check records whether a declared field changed to its expected value on at least one complete repeat. All repeat-level matches, contradictions, and unknowns remain visible. A semantic expectation is not automatically verified, and a changed recorded field is not proof of a correct causal explanation or improved overall judgment.

**Candidate acceptance and program inheritance.** Candidate acceptance requires complete, comparable execution, a strict development-agreement gain over the best evaluator, nondecreasing coverage, and the prespecified worst-stratum guard. These are development selection rules rather than statistical significance claims. Program inheritance is a separate decision. Inheritance first requires known, complete, comparable local and development outcomes, including comparisons against both the parent and best program. Subject to that requirement, accepted candidates and candidates with no observed damage can be inherited. For this decision, damage includes a local ordering regression, deterioration on a declared protection case, or more regressed than corrected sealed-Train comparisons against either the parent or the best evaluator. An unaccepted candidate with damage is eligible only when fresh measurement supplies new progress and the declared protection cases do not deteriorate.

Qualifying progress is either a first-time repair of an error present at the direction’s root, still wrong in the parent, or a newly observed, predeclared exact intermediate-field change. Previously credited repairs and checks are retained across attempts, so breaking and repairing an already corrected case cannot renew this allowance. A missing protected case is unknown, not preserved. When inheritance fails, the next parent returns to the pre-update best program, even if the candidate has just been accepted as the new best. The candidate and its diagnostics remain available. Keeping an intermediate candidate never changes the recorded best score.

**Direction continuation and termination.** In the native editing session, the agent can request diagnostics, submit a candidate, repair an invalid submission, or explicitly pause with its unresolved question. The controller enforces parent eligibility, validity, and resource limits; a failed inheritance check returns the editable program to the pre-update best without erasing the direction’s history. The next editing opportunity remains within the direction until a pause or an execution/resource limit ends that allocation. The controller does not automatically judge whether a direction is unhelpful, and the native workflow does not require a separate model reviewer to certify direction continuation. That judgment requires diagnostic results contradicting the hypothesis after checking for missing observations, unused tool output, or incorrect interpretation. One unsuccessful implementation, a nonsignificant difference, or budget exhaustion alone is insufficient. These requirements govern interpretation of diagnostic results, not an automatic hypothesis test.

**Terminal assessment and attribution.** The terminal shortlist includes eligible, previously accepted, complete programs with positive development gain over the seed and is ordered using Train evidence alone. Its membership and order are frozen before Val access. Terminal Val does not rerank the shortlist or guide another proposal. With no eligible improvement, the seed is retained without an additional Val query. Fresh system confirmation and independent Test/OOD assessment remain separate from adaptive development.

A connected candidate receives one system-level outcome, not independent causal credit for each component. Component attribution requires paired interventions with the remaining interfaces fixed,

such as tool on/off or evolved versus seed synthesis. If removing a dependency makes the program invalid, the comparison must report that limitation rather than substitute an unrelated whole-program control. Complete prompt templates, interaction traces, candidate changes, and execution settings belong in the reproducibility record rather than the main algorithm.

## E SEMANTIC SYNTHESIS AND ANCHORED SCORING

The default fusion procedure forms an overall judgment from specialist findings; a declared numerical rule is an explicit alternative. Each specialist produces findings with identifiers, supporting evidence, dimensions, and uncertainty, as well as its own rating and reason. The full specialist trace remains available for diagnosis, but the synthesizer’s default input omits the structured specialist score, rating, and score scale. It retains the findings, reasons, and scope of each specialist and applies its own overall scale. This input separation does not remove value judgments expressed in the reasons.

The synthesizer groups findings by their consequences for quality, including strengths, weaknesses, mixed effects, and uncertainty. Every finding reference must occur in exactly one group. This accounting rule makes omitted or multiply referenced findings detectable; it does not guarantee correct semantic grouping. Duplicate descriptions should not become independent penalties, and favorable style should not cancel a factual error when the domain standard prioritizes faithfulness.

Let  $a_k < a_{k+1}$  be adjacent overall anchors and  $\lambda \in [0, 1]$  locate quality between them. The resulting scalar is

$$s = (1 - \lambda)a_k + \lambda a_{k+1}. \quad (8)$$

The anchors have domain-specific verbal meanings; specialist scores with different meanings are not averaged. Reference completeness, semantic correctness, and calibration require separate checks. A synthetic arithmetic example uses  $a_k = 50$ ,  $a_{k+1} = 75$ , and  $\lambda = 0.4$ , giving  $s = 60$ . These numbers illustrate interpolation, not a universal scale or an evaluator result.

## F MEASUREMENT AND STATISTICAL ANALYSIS

**Completeness and denominators.** Use the original-input weights  $w_{gj} = 1/(|S||P_g|)$  from Equation 5, fixed before execution. Let  $c_{gj} = 1$  when both artifacts have all  $R$  valid complete scores,  $M = \sum_{g,j} w_{gj} c_{gj}$  be covered weight, and  $K = \sum_{g,j} w_{gj} c_{gj} z_{gj}$  be correct weight. Report conditional agreement  $K/M$  when  $M > 0$ , weighted coverage  $M$ , raw completion counts, and the missing-outcome interval

$$[K, K + 1 - M]. \quad (9)$$

For  $n$  equally weighted pairs with  $m$  complete and  $k$  correct, this reduces to  $[k/n, (k + n - m)/n]$ . The interval bounds full-cohort agreement under arbitrary missing outcomes; it is not a statistical confidence interval. A missing repeat is not replaced by the median of surviving calls. Completeness includes selected routes and tools, not only a final score. A valid observation that an artifact fails a quality check differs from an execution error.

**Ranking and selection endpoints.** For the default  $N = 4$  design, human complete rankings support all strict pairs. Global agreement averages within input groups before aggregation across inputs. Top-region agreement uses prespecified strict comparisons involving at least one human top-two output. Best-of- $N$  measures selection of a human top-ranked output; normalized rank regret is defined in Section 6.1. Human ties, evaluator ties, and incomplete rankings are governed by a frozen policy and reported separately. Pair-only data do not supply missing ranks or cardinal utility. Core  $H_1$  agreement is reported by dimension before any macro average. Groups without a strict human preference are ineligible for strict agreement and retain a separate count; they are not assigned artificial pair labels. Repeated appearances of an artifact use its frozen scores, so pairs sharing an input or artifact are not independent observations.

**Randomness and repeated use.** Freeze  $R \geq 3$ , score precision, retry policy, artifact reuse, and cache identity. The target in Equation 4 averages over this fixed execution protocol as well as the input distribution; a finite set of recorded executions is one realization, not the population median.

Retries are outcome-blind and uniformly applied. Repeated scores are not independent task observations. Paired uncertainty uses original-input/source group resampling, preserving output groups and comparisons; the planned interval level is 95%. Report independent search seeds separately from scoring repeats, and distinguish within-run execution variation from between-search variation. Retain failed, rejected, incomplete, and interrupted attempts in the authorized analysis, without favorable resampling or denominator changes.

**C2/C3 human outcomes.** Report wins, losses, ties, invalid outputs, and major regressions per input group. A prespecified tie-aware preference probability may assign half credit to a tie, with its denominator and missingness bounds stated. Report C2 changes against the shared  $y_0$  in addition to the critic contrast, and C3 changes against Base in addition to the reward contrast. Intervals and costs accompany effect sizes. Multiple domains, dimensions, and ablations require declared primary contrasts and a prespecified multiplicity policy. A search-selected development maximum is not substituted for any final endpoint.

## G PROGRAM EVOLUTION AND PARAMETER ADAPTATION

Parameter adaptation is a competing approach to learning human evaluation standards, not an assumed failure case. Table 6 distinguishes score calibration, evaluator fine-tuning, and program evolution by their editable objects. Fine-tuning can learn criteria and tool-use behavior when its training interface permits them. The fixed-structure control below is a declared experimental restriction, not a claim about the limits of fine-tuning in general.

Table 6: Adaptation choices for the planned C1 comparison. Rows specify what changes, not a ranking of effectiveness. All use the same permitted human feedback; observed evidence and total adaptation cost must also be reported.

| Approach              | Adapted object                                                                                          | What the comparison tests                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Score calibration     | Combination weights, scale maps, or thresholds over frozen seed signals; model and evidence path fixed  | Whether the preference mismatch can be addressed by a different numerical decision rule.  |
| Evaluator fine-tuning | Judge weights from the common base checkpoint; seed adapters, tools, routes, and output interface fixed | Whether learning a new mapping from the same observations to judgments is sufficient.     |
| IterEval              | Artifact adapters, criteria, evaluator modules, tools, routing, and fusion; base-model weights fixed    | Whether changing the evaluation procedure improves alignment beyond parameter adaptation. |

**Common human feedback and selection.** All arms use the same proposal-Train inputs, artifacts, and available human feedback levels. A fine-tuned evaluator learns pointwise scores from the permitted comparisons, with dimensional feedback when available; missing human explanations are not replaced by extra teacher labels in this contrast. Scoring remains pointwise and label-blind. Sealed Train supplies aggregate measurements only, never per-example training targets or gradients. Tuning configurations and checkpoints follow the same terminal-Val and locked-Test/OOD boundary as program candidates. Human units, annotation time, unsuccessful trials, optimization compute, tool calls, and selection costs are all counted. Adaptation and deployment costs are reported separately, with matched-budget comparisons wherever feasible and unmatched resources explicitly identified.

**Separating observation from interpretation.** The end-to-end comparison permits IterEval to change its observation path. It cannot by itself attribute a difference to better interpretation of identical evidence. A shared-observation control therefore freezes  $(A, T, \Pi)$  and supplies the same artifact evidence to fine-tuning and IterEval restricted to  $(K, V, G)$ . This control, together with the component comparisons in Table 11, tests whether changed observations account for any advantage.

Table 7: Required empirical outputs beyond the simulated reporting examples. Every result requires authorized frozen data and a traceable analysis; neither conceptual illustrations nor simulated displays supply such evidence.

| Output           | Required summary                                                                                                            | Interpretation limit                                                               |
|------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| C1 alignment     | Global/top-region agreement, per-dimension $H_1$ , worst stratum, ID/OOD, coverage, repeated-score variation                | Pairwise improvement does not establish selection or generation gains.             |
| Feedback quality | Label origin, source protocol, annotator pool, agreement, abstention, unresolved judgments, and annotation time per setting | Newly collected labels are not automatically more reliable.                        |
| C1 selection     | Best-of- $N$ , normalized rank regret, complete human rankings, grouped intervals                                           | Rank regret is not cardinal utility; pair-only corpora are insufficient.           |
| C2 refinement    | New blinded contrast, changes against $y_0$ , major regressions, dimensional quality, external metrics                      | Fresh shared initial outputs and matched refinement are necessary.                 |
| C3 training      | New blinded reward contrast, Base sanity check, external metrics, validity, reward-hacking diagnostics                      | Design-only until separately authorized training and final evaluation occur.       |
| Efficiency       | Human units and time, search calls/tokens/tool use, runtime, deployment and training costs                                  | More compute or annotation detail is not a free method gain.                       |
| Trajectory       | Directions, implementation attempts, continuation/pause, unresolved stops, resource allocation                              | A schematic tree is not an optimization trace or evidence of efficiency.           |
| Mechanism        | Paired component effects, interactions, remaining gap to human agreement, scoped counterexamples                            | Assign limitation regimes only after evidence; do not generalize from one stratum. |
| Adapter fidelity | Direct/derived label agreement and downstream differences on the same subset                                                | Annotation conversion cannot change the construct being evaluated.                 |

Neither lower annotation cost nor superiority to fine-tuning is assumed. A combined weight-and-program arm would be needed to test complementarity.

**Inspectable does not mean faithful.** An explicit rubric and evidence trace make the stated basis of a judgment available for inspection. They do not establish that the judgment follows that basis or matches the human standard. Mechanism assessment must check cited observations and test predicted decision changes under controlled evidence or component interventions. These tests are distinct from preference agreement.

The tuning comparison concerns the *evaluator*. C3 instead trains a *generator* against frozen evaluator rewards. Neither experiment has been executed in this draft. The simulated C1 displays illustrate a subset of reporting conditions and contain no fine-tuning or calibration outcomes.

## H DETAILED CORE RESULTS AND EXTERNAL ENDPOINTS

Empirical plots for the outcomes in Table 7 require frozen reporting cohorts and analysis. Suitable outputs include human-budget curves, per-domain component effects, annotated search trajectories, and C2/C3 regression breakdowns. The numerical displays in Appendix L are reporting fixtures, not provisional substitutes for those measurements. Table 8 records the empirical status of the downstream studies independently of the simulated C2 illustration.

**C1 disaggregation.** Each populated main-table result will be accompanied by domain, feedback origin, split, original-input count, output count, and completion rate. Within the same registered cohort, report overall and dimensional agreement, top-region agreement, selection accuracy, rank regret, worst-stratum agreement, and repeated-score variation. Report equal-domain summaries separately for reused and new preferences, with their paired IterEval-minus-baseline effects. A main-



table summary uses equal-domain weights and a common registered domain set across methods for each endpoint. Report the eligible-domain count for each endpoint; pair-only domains cannot silently change the domain set used for selection metrics. Missing domains remain visible rather than being removed according to performance. Pair-only corpora contribute only to the pairwise endpoint. Per-domain plots replace the simulated fixtures only after the underlying runs and grouped intervals are verified. Table 1 is conditional on a fixed, preregistered search replicate rather than the best replicate selected after evaluation. Its paired intervals resample original-input/source groups within domains with the evaluators fixed. Report independent-search means, dispersion, and run counts alongside this fixed-evaluator comparison; scoring repeats do not increase the independent-search count.

**Table 8: Downstream human outcomes: unmeasured placeholders.** C2 compares IterEval-refined with Seed-refined; C3 compares IterEval-GRPO with Seed-GRPO. Preference cells await tie-aware estimates, input-group 95% intervals, and counts. Each  $\Delta\text{Reg}$  cell is the IterEval-minus-seed difference in major-regression rate against  $y_0$  (C2) or Base (C3), with ID and OOD separate. Appendix H specifies per-arm rates and completion. Dashes await new blinded judgments; C3 is design-only.

| Study | Domain / role                | ID pref. $\uparrow$ | OOD pref. $\uparrow$ | ID $\Delta\text{Reg}$ $\downarrow$ | OOD $\Delta\text{Reg}$ $\downarrow$ |
|-------|------------------------------|---------------------|----------------------|------------------------------------|-------------------------------------|
| C2    | SV4: specification-to-web    | —                   | —                    | —                                  | —                                   |
|       | IG1: detailed image captions | —                   | —                    | —                                  | —                                   |
|       | TG1: summarization           | —                   | —                    | —                                  | —                                   |
| C3    | IG1: primary candidate       | —                   | —                    | —                                  | —                                   |
|       | SV2: second candidate        | —                   | —                    | —                                  | —                                   |
|       | TG4: verifiable control      | —                   | —                    | —                                  | —                                   |

**C2 output comparisons.** For each input, retain the fresh  $y_0$ , Seed-refined output, and IterEval-refined output as one blinded ranking group. Report the IterEval-versus-seed preference with ties and abstentions separate, each arm’s improvement and regression relative to  $y_0$ , major regressions, and core  $H_1$  changes. For Table 8, each regression contrast subtracts the seed arm’s major-regression rate from the IterEval arm’s rate within one domain and split. The C3 contrast uses the same definition relative to Base, rather than relative to a C2 starting output. Per-arm rates, paired uncertainty, and missing-outcome bounds accompany both contrasts. The generator, critique interface, observation access, decoding, and revision budget remain identical across arms. Benchmark outputs can support pilots, but do not replace fresh generation in the confirmatory comparison. IG2 and AR1 enter only after new human or expert assessment is ready; their absence is not evidence of failure.

**Table 9: External endpoints for C1–C3, separate from human preference.** All outcome cells are unmeasured. Specific metric versions and applicability must be frozen before final evaluation; no average is taken across different quality constructs. Reserved checks never become mining signals, evaluator tools, critics, or rewards.

| Domain              | Targeted attribute           | Checks and protected attributes                                                       | Outcome |
|---------------------|------------------------------|---------------------------------------------------------------------------------------|---------|
| SV4                 | Specification satisfaction   | Functional checks; visual usability and no lost required content                      | —       |
| IG1                 | Caption grounding and detail | Factuality, hallucination, and content coverage; length alone is insufficient         | —       |
| TG1                 | Summary faithfulness         | Source support and key-content coverage; protect against omission                     | —       |
| SV2                 | Reconstruction fidelity      | Render comparison and executability; visual similarity alone is insufficient          | —       |
| TG4                 | Verifiable quality control   | Applicable reference or executable checks; do not generalize to every open-ended task | —       |
| Other ready domains | Domain-specific standard     | Report the applicable reserved check, or explicitly record no suitable metric         | —       |

Table 9 complements human outcomes rather than replacing them. When the two disagree, inspect the same outputs, dimensional judgments, and metric scope before attributing the difference to reward exploitation or metric failure. Distinguish an unrun experiment, an inapplicable metric, and a completed experiment without confirmed improvement.

## I PLANNED C3 TRAINING PROTOCOL

C3 is specified to test reward usefulness independently of evaluator agreement and critic usefulness. It is not an executed experiment, and no parameter training is part of this manuscript preparation.

**Representative tasks and pre-training diagnostics.** IG1 is the primary training candidate, SV2 a possible second task, and TG4 a verifiable control; AR1 is not a priority for the first training study. Before training, score a common diagnostic output pool from the actual target policy with both frozen evaluators. Check top-region preferences against new human judgments, repeated reward variance, ties, invalid scores, and sensitivity to length, omission, and format. Only this preflight diagnostic pool is shared: after updates begin, each policy generates its own online rollouts. The reward version, score mapping, tie and invalid-output policy, and Train/Val/Test separation must be frozen before policy updates. A domain proceeds only after prespecified stability and exploitation checks pass.

Table 10: Controls for the planned LoRA + GRPO comparison. Numerical hyperparameters, pilot thresholds, and final sample sizes remain to be frozen in the experiment manifest.

| Control             | Contract                                                                                                                                      |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Arms                | Base is untrained; Seed-GRPO and IterEval-GRPO start from the identical base checkpoint.                                                      |
| Training data       | Same unlabeled training inputs and order; source-level deduplication and contamination audit.                                                 |
| Policy updates      | Same LoRA structure, optimizer, learning rate, update count, group size, KL rule, checkpoint cadence, and policy-training compute.            |
| Online samples      | Each updated policy samples its own rollout groups. Do not share trajectories once policies diverge.                                          |
| Reward version      | Freeze program, prompt, tools, routing, aggregation, normalization, ties, abstention, and invalid-output policy before training.              |
| Reward resources    | Equal reward-call counts and token caps; report actual inference time, tool use, and monetary or compute cost separately.                     |
| Training validation | Input-disjoint development validation selects checkpoints and checks reward exploitation; final metrics and final human labels remain locked. |
| Pilot gate          | Prespecified acceptable validity, reward variance, KL, length, stability, held-out quality, and reward-hacking diagnostics before expansion.  |
| Final generation    | Freeze checkpoints and decoding, then generate new Base, Seed-GRPO, and IterEval-GRPO outputs on the same unseen ID/OOD inputs.               |
| Final judgment      | New blinded $H_0$ and core $H_1$ , reserved external metrics, invalid/major-regression rates, length and format shifts, and total cost.       |

Under the controls in Table 10, the frozen reward produces scalar scores within each rollout group using the same normalization and advantage rule. Zero-variance groups, invalid artifacts, and execution failures require a common prespecified policy. Deterministic checks may enforce format or executability, but do not define the entire reward for open-ended quality.

The pilot uses development validation only. If its prespecified stability or reward-hacking gate fails, expansion stops for that domain and the failure is retained. Success elsewhere is not a substitute. Final sample sizes are determined from pilot variance and a declared minimum detectable effect before test generation. Test inputs and judgments do not participate in training, checkpoint selection, reward design, or failure-driven edits.

Any deliberate sharing of locked input pools across C1/C2/C3 must be prespecified with an access audit; their newly generated outputs and judgments are never treated as reusable old labels.

## J FULL ABLATION MATRIX

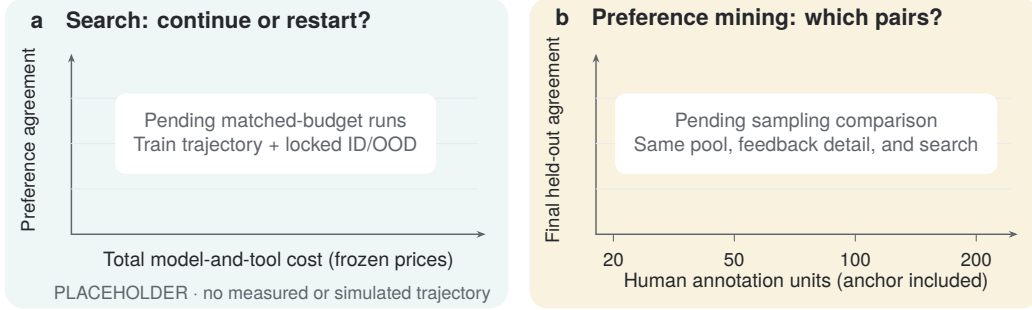


Figure 5: **Matched-budget comparisons awaiting measurements.** (a) Search progress under equal total model-and-tool cost at frozen prices; locked ID/OOD checkpoint assessment occurs after search and never guides it. (b) Final agreement under equal human budgets and a fixed evolution budget. No curves or gains are supplied. Table 11 specifies the controls.

**C1 comparator contract.** The rows in Table 1 are controlled comparison specifications, not claims that every baseline configuration has been executed. The best single metric is selected per domain using Train only from a prespecified applicable metric set, with score orientation fixed before selection. The metric ensemble uses fixed normalized weights declared before human labels are inspected; fitting these weights belongs to the score-calibration arm. The static model judge uses a fixed prompt and no added tools, while the static tool judge is the complete unevolved seed. Score calibration and evaluator fine-tuning follow Appendix G. Evaluator fine-tuning changes model weights; score calibration changes numerical mappings over frozen seed signals. Prompt optimization uses the history-only allocation defined below, restricted to instructions in  $(K, V, G)$  with adapters, tools, and routes fixed. Program search uses that allocation with the same full editable scope as IterEval. These internal controls do not stand for all implementations of GEPA, SkillOpt, or other published optimizers. Exact model/checkpoint versions, metric sets and normalizations, tuning configurations, and monetary caps remain pending in Appendix M and must be frozen before runs.

**Search allocation contract.** All four conditions use the same initial program, proposal-Train pool, agent model, edit interface, permitted diagnostics, acceptance guard, and terminal selection rule. Independent proposals always edit the current best program and receive no previous failed-attempt diagnostics. History-only proposals also edit the current best program, but can inspect their own run’s accumulated diagnostic history under the same access rules as IterEval, without an active direction receiving priority or an unaccepted program becoming the parent. Depth-first allocation keeps the active direction and the program-inheritance rule of Section 5.3, without returning to a paused direction. The pause/resume extension additionally allows reopening a saved direction and its history, subject to the same eligibility rules. Its reopening policy remains to be frozen, and the row is design-only. The history-only versus depth-first contrast tests the combined allocation and inheritance policy, not either mechanism in isolation. Attributing a gain specifically to inheritance would additionally require depth-first allocation with the parent reset to the best program after every rejection.

The primary search budget is total model-and-tool cost evaluated under one frozen price schedule. It includes proposal tokens, requested diagnostics, failed calls, candidate measurement, and a reserved terminal-assessment allowance. All arms stop at the same cap; differences in numbers of attempts are outcomes, not a second matching requirement. Token counts, tool calls, elapsed time, and attempts are reported separately. Fixed post-search Test/OOD evaluation costs are also reported, but never buy additional search or select checkpoints. Fine-tuning requires a separate accounting of training compute; a common monetary cap alone does not make unavailable checkpoint or observation access comparable.

The comparisons in Table 11 hold nonintervened components and evidence access fixed under the declared resource contract. If method interfaces differ, report that difference instead of attribut-

Table 11: Planned ablation conditions and the inference each comparison supports. All conditions require matched evidence access and accounted computation. An oracle condition is diagnostic, not a deployable comparator.

| Factor              | Conditions                                                                                                        | Endpoint or interpretation                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Data mining         | Random; judge uncertainty; cross-model conflict; run instability; Disagreement-based preference mining            | Final held-out alignment at equal $N_H \in \{20, 50, 100, 200\}$ ; the number of selected disagreements alone is insufficient. |
| Human detail        | Same pairs with $H_0$ , $H_0 + H_1$ , $H_0 + H_1 + H_2$ , and $H_0 + H_1 + H_2 + H_3$                             | Marginal value of granularity; report annotation units and annotation time.                                                    |
| Editable scope      | Full $(A, K, V, T, \Pi, G)$ ; rubric/procedure only $(K, V)$ ; fixed tools and static routes; routing only        | Whether additional observations or structural changes are needed beyond criteria changes.                                      |
| Evidence ceiling    | Oracle evidence and/or oracle routing with access declared                                                        | Diagnostic headroom under privileged information; no deployment claim.                                                         |
| Search allocation   | Independent one-shot proposals; history-only proposals; direction depth-first; direction search with pause/resume | Value of direction management at equal total search and measurement budgets.                                                   |
| Tool activation     | Tool on/off with compatible consumers                                                                             | Contribution of acquiring the observation, not the whole program bundle.                                                       |
| Tool implementation | Evolved tool versus seed tool                                                                                     | Contribution of changed measurement or evidence extraction.                                                                    |
| Evidence route      | Adaptive versus fixed routing with the same tool set                                                              | Contribution of which observations reach which modules.                                                                        |
| Rubric and modules  | Evolved versus fixed criteria/procedures                                                                          | Contribution of judgment changes with observations held fixed.                                                                 |
| Fusion              | Declared numerical rule versus semantic synthesis                                                                 | Contribution of resolving priorities, overlap, and conflicting findings.                                                       |
| Feedback adapter    | Direct pairs versus derived ranking/pointwise/error labels on the same subset                                     | Label fidelity and its downstream effect; keep original input dependence.                                                      |

ing an advantage to search allocation. Component attribution uses paired original-input groups and prespecified compatible replacements; interactions may prevent an independent per-component interpretation.

Table 12: Matched-budget search comparisons: unmeasured placeholders. All strategies share feedback, seed, editable components, and measurement protocol. Report independent-search means, dispersion, and run counts separately from input-group intervals for each frozen evaluator; account for all diagnostic and failed calls. The pause/resume row is a planned condition, not a claim of completed implementation or evaluation.

| Search allocation          | ID | OOD | Attempts/direction | Total cost |
|----------------------------|----|-----|--------------------|------------|
| Independent proposals      | —  | —   | —                  | —          |
| History-only proposals     | —  | —   | —                  | —          |
| Direction depth-first      | —  | —   | —                  | —          |
| Depth-first + pause/resume | —  | —   | —                  | —          |

Table 12 contrasts history access, the combined depth-first policy, and the planned addition of pause/resume. The trajectory analysis records the cost before the first gain, later best-so-far gains, and the fate of each direction. Frozen-checkpoint ID/OOD assessment occurs only after search; it cannot select the next attempt. Independent searches, rather than repeated scoring of one final evaluator, quantify optimizer variability.

**Existing-dataset mining replay.** Existing output groups remain available to the miner, but their human labels are hidden until the sampling policy selects them. Freeze source mappings and the random anchor before targeted selection; no final label or mining-conflict score enters the deployed

evaluator. Use the same original-input annotation units, feedback detail, and evolution budget for every policy. Replay tests label efficiency, not newly incurred human effort. Use historical annotation time only when compatible time records exist; otherwise record it as unavailable. For new annotation, report agreement against both units and measured human time, including extra raters, adjudication, and abstentions. The random arm’s anchor is part of its uniformly sampled budget, not an additional set of labels.

## K TRAJECTORY, MECHANISM, AND EFFICIENCY ANALYSIS

**From unsuccessful implementation to accepted change.** For each selected trajectory, report the direction’s initial error, hypothesis, candidate sequence, diagnostic measurements, and acceptance and inheritance decisions. A favorable example must show what changed between the failed and successful attempts, not merely that the later score is higher. Include unsuccessful and exhausted directions under the same selection rule. Judging a direction unhelpful requires a contradicted prediction and checks of relevant implementation problems; the controller does not make this judgment automatically.

**Controlled component attribution.** Compare compatible tool, route, rubric, module, and synthesis variants on the same inputs. Hold upstream observations fixed when testing their interpretation; rerun the full program when the observation path changes. A tool can matter only if its output reaches a consumer and changes the resulting judgment. Interactions between components prevent interpreting full-minus-fixed effects as additive contributions. Only after these tests should a failure be assigned to missing evidence, incorrect interpretation, or insufficient feedback.

**Feedback and adapter fidelity.** Assess  $H_0$  through  $H_3$  cumulatively on the same pairs and include annotation time. Direct and converted preferences are compared on the same original-input subset, first for label fidelity and then for downstream agreement. A label conversion does not create missing human rationales or dimensional judgments. Agreement among existing signals and agreement among annotators are distinct measurements.

**Efficiency and deployment.** Report data-mining computation, human time, search cost, and deployment cost separately. Search cost includes rejected candidates, diagnostic calls, retries, and terminal assessment. Repeated pointwise scoring measures execution variation; independent search seeds measure optimizer variation. A cost–agreement profile is not an evolution trajectory, and a longer direction does not by itself establish efficient search.

## L SIMULATED REPORTING FIXTURES

**Main-text C1 layout.** Figure 3 retains the landscape layout but uses all 3,520 requested OOD input groups in the existing synthetic records, not the 2,880-input subset in Figure 6. Within each task, pairwise agreement uses the intersection of complete inputs across the five displayed methods; all five bars therefore share the same denominator. The intervals resample original inputs jointly across methods with 2,000 bootstrap draws, rather than treating pairs or scoring repeats as independent observations. Panel (b) averages these task means with equal weights within each family. Its selection endpoint uses all 160 requested inputs per task, counting failed scoring as failed Best-of-4 selection. The different endpoint denominators are recorded in `data/simulated/c1_landscape/`; no task is selected or omitted according to its outcomes. Connecting seed and IterEval points identifies the same family, not a search trajectory or a causal effect. The four equal ring quadrants represent the task taxonomy, not measured sample proportions. The figure does not simulate calibration, fine-tuning, sampling efficiency, search progress, C2, or C3. The main table retains the full comparator specification, and no empirical claim is supported by this layout fixture.

**Purpose and limits.** The quantitative figures in this appendix and Table 14 are generated from explicitly simulated data for reviewing the reporting design. The generator does not execute IterEval, run a model, consult benchmark outcomes, or recruit annotators. Its assigned noise, effects, costs, and cohort labels are not estimates of expected performance, a power analysis, or evidence for method ranking. Positive, near-zero, and negative contrasts are retained. None of these fixtures

Table 13: Eight diagnostic cohorts for the simulated C2, annotation, and component displays. The C1 displays cover all 22 domains separately. Assignments are layout examples, not verified data availability or fixed domain categories. Each C1 domain/split contains 160 synthetic input groups and four outputs.

| Family             | Reused-preference example             | New-preference example                  |
|--------------------|---------------------------------------|-----------------------------------------|
| Structured visuals | TD-SV4: specification-conditioned web | TD-SV2: webpage reconstruction          |
| Images / 3D        | TD-IG1: detailed image captioning     | TD-IG3: instruction-based image editing |
| Text generation    | TD-TG1: document summarization        | TD-TG2: RAG-supported answers           |
| Automated research | TD-AR1: research ideas                | TD-AR2: ablation plans                  |

provides evidence for sampling efficiency, component effectiveness, or human-label reliability. No C3 training outcomes are simulated.

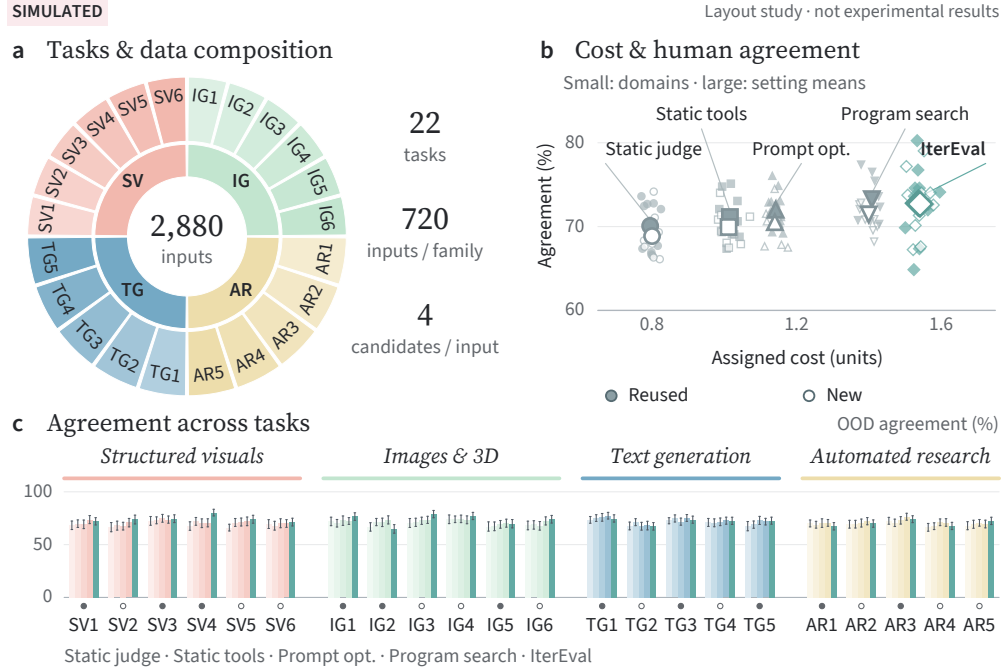
**Construction and uncertainty.** The C1 fixtures cover all 22 domains in Table 2, with 11 illustrative cohorts per feedback setting. Table 13 retains eight fixed tasks for C2 and diagnostic detail; it does not define the study’s scope. A domain may use both feedback settings in an actual study. Gaussian latent utilities define synthetic C1 rankings. Noisy scores are repeated three times per artifact and reduced to medians before strict comparison. Incomplete groups remain in coverage denominators. The C1 table averages domain-level metrics equally within each setting. Its means use each method’s complete groups, whereas the paired differences in Figure 7 use the intersection of completed groups. Intervals use 1,000 input-group bootstrap replicates of these artificial observations. They quantify resampling variation in the fixtures, not uncertainty about IterEval. The sampling-fixture generator assigns budget responses for five policies and 12 toy runs; its bands are percentile ranges across runs, not bootstrap intervals.

**Balanced subset in the landscape figure.** Figure 6 uses a subset of the OOD fixtures, with 720 input groups per family. The six-task families contain 120 inputs per task; the five-task families contain 144. Within each task, input identifiers are ranked by SHA256 with fixed seed 20260918, without reference to outcomes. The same selected inputs, including incomplete responses, are retained for every method. Agreement and its 1,000-replicate bootstrap intervals are recomputed on each method’s complete inputs; cost retains all selected inputs. Setting means weight their 11 domains equally, rather than pooling inputs. The ring’s equal quarters reflect this synthetic allocation, not measured domain prevalence or a proposed empirical sampling rule. The released selection list and numerical exports distinguish this display from the full-cohort tables and appendix plots.

**Expanded alignment and sampling displays.** Figure 7a–b reports per-domain IterEval-minus-seed effects with paired intervals for all 22 domains on the full synthetic cohorts. These paired contrasts complement the marginal means for the balanced OOD subset in Figure 6c. Figure 8a–b compares sampling policies against random sampling at equal human budgets; all policies include the same reliability anchor. Annotation granularity is held fixed here and varied separately in Figure 11b.

**Annotation outcomes.** The annotation fixture assigns three simulated raters per group. Preferences for the two outputs are encoded as 0/1, a tie as 2, and abstention as −1. Agreement is computed within each group from all available non-abstaining rater pairs, then averaged across eligible groups. Fewer than two valid votes is insufficient; three distinct valid votes yields no consensus. A two-of-three majority gives a decisive preference or a tie. These outcomes are not evidence about actual annotators or any public benchmark.

**Selection beyond pairwise agreement.** Figure 10a–b reports the entire selected-rank distribution. Rank 1 is the synthetic human optimum; unscored inputs remain visible. This full-input denominator differs from the completed-group Best-of-4 accuracy in Table 14. Both definitions are exported with their denominators so missingness cannot masquerade as better selection.



**Figure 6: Simulated task coverage and evaluator comparisons; not experimental results.** Panel (a) summarizes 22 tasks and 2,880 input groups, balanced to 720 per family with four outputs per input. A fixed hash selects this subset without reading outcomes; the appendix reports the full synthetic cohorts. Sector angles encode input counts; task codes follow Table 2. Panel (b) plots assigned cost units, not measured runtime, against OOD agreement with synthetic human rankings. Small marks denote 110 domain–method combinations; large marks equally average 11 domains per feedback setting. Shapes identify methods, and filled/open marks denote illustrative reused/new preferences, not label reliability. Panel (c) reports agreement from pointwise scores and 95% input-group bootstrap intervals on a common 0–100% scale. Baseline bars follow the displayed order in each family’s hue; IterEval is lake green. Cost includes all selected inputs; agreement uses method-complete inputs. Marginal intervals do not establish paired differences.

## SIMULATED DATA

NOT EXPERIMENTAL RESULTS

## 22 DOMAINS / 4 FAMILIES

## Structured visuals · 6

|     |                         |   |
|-----|-------------------------|---|
| SV1 | Brief → single slide    | R |
| SV2 | Screenshot → webpage    | N |
| SV3 | Data → visualization    | R |
| SV4 | Specification → webpage | R |
| SV5 | Sources → slide deck    | N |
| SV6 | Topic → slide deck      | N |

## Images / 3D · 6

|     |                           |   |
|-----|---------------------------|---|
| IG1 | Detailed captioning       | R |
| IG2 | Text → image              | R |
| IG3 | Instruction-based editing | N |
| IG4 | Super-resolution          | N |
| IG5 | Text → 3D                 | R |
| IG6 | Image → 3D                | N |

## Text generation · 5

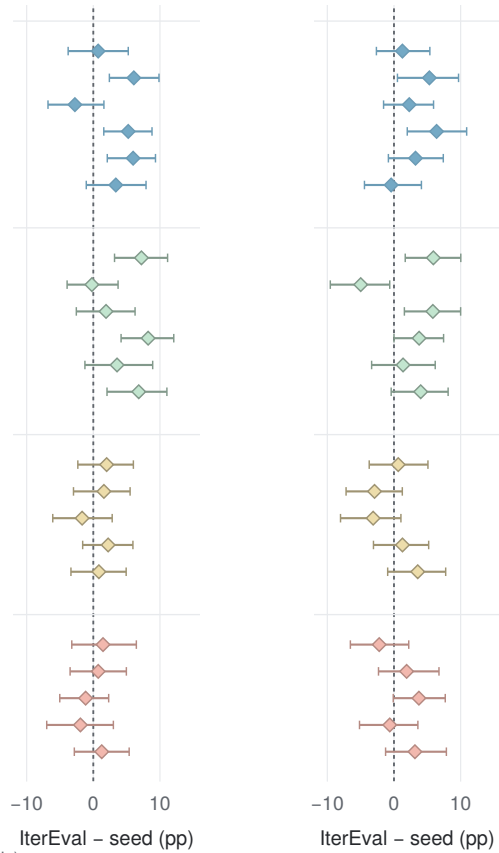
|     |                         |   |
|-----|-------------------------|---|
| TG1 | Document summarization  | R |
| TG2 | RAG long-form answers   | N |
| TG3 | Translation             | R |
| TG4 | Instruction constraints | N |
| TG5 | Open dialogue           | R |

## Automated research · 5

|     |                      |   |
|-----|----------------------|---|
| AR1 | Research ideas       | R |
| AR2 | Ablation plans       | N |
| AR3 | Table → insight      | R |
| AR4 | Related work         | N |
| AR5 | Peer-review critique | N |

a ID test

b OOD test



R: reused / N: new preferences (illustrative assignments only).  
 Diamonds: paired means. Whiskers: 95% bootstrap intervals over synthetic input groups.

Figure 7: **Simulated domain-level alignment; not experimental results.** Panels (a) and (b) compare IterEval with the static tool-augmented seed on ID and OOD inputs. Diamonds are paired mean differences; whiskers are 95% bootstrap intervals over jointly completed synthetic input groups. Zero denotes no difference. All 22 domains retain the 6/6/5/5 taxonomy order. Colors identify domain families; R/N denote illustrative reused/new feedback assignments, not permanent domain categories or measured data readiness.

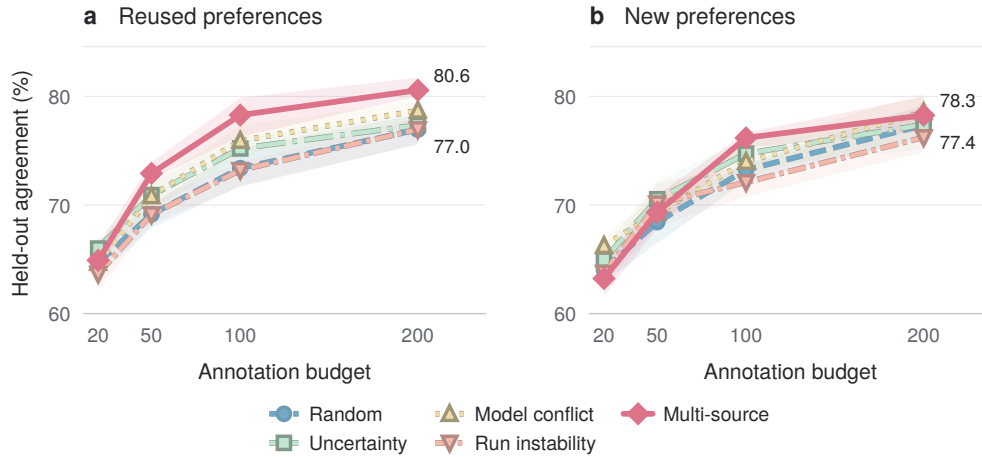


Table 14: **Simulated C1 reporting example; not experimental results.** Each setting averages eleven equally weighted toy cohorts. Pair = pairwise agreement; Top = top-region agreement; Bo4 = Best-of-4 accuracy (all in %); Reg = normalized rank regret (lower is better). Cohort assignments do not assert actual label availability. Coverage and domain summaries accompany the reproducible fixtures.

| Evaluator                           | ID test |      |      |      | OOD test |      |      |      |
|-------------------------------------|---------|------|------|------|----------|------|------|------|
|                                     | Pair    | Top  | Bo4  | Reg  | Pair     | Top  | Bo4  | Reg  |
| <b>SIMULATED</b> Reused preferences |         |      |      |      |          |      |      |      |
| Single metric                       | 66.5    | 67.5 | 44.8 | 0.31 | 64.1     | 65.2 | 43.5 | 0.32 |
| Metric ensemble                     | 69.9    | 70.9 | 51.2 | 0.26 | 66.8     | 68.0 | 45.3 | 0.30 |
| Static judge                        | 71.3    | 72.3 | 52.2 | 0.25 | 69.9     | 71.1 | 49.8 | 0.27 |
| Static tools                        | 74.0    | 75.2 | 57.1 | 0.23 | 71.3     | 72.9 | 52.8 | 0.24 |
| Prompt optimization                 | 74.1    | 75.5 | 56.8 | 0.22 | 71.9     | 72.9 | 54.4 | 0.24 |
| Program search                      | 74.8    | 76.1 | 57.7 | 0.21 | 73.0     | 74.5 | 55.5 | 0.23 |
| IterEval                            | 75.3    | 76.9 | 61.2 | 0.20 | 72.7     | 74.2 | 55.3 | 0.23 |
| <b>SIMULATED</b> New preferences    |         |      |      |      |          |      |      |      |
| Single metric                       | 66.2    | 67.0 | 45.1 | 0.30 | 63.3     | 64.3 | 40.5 | 0.34 |
| Metric ensemble                     | 68.3    | 69.6 | 47.8 | 0.29 | 65.9     | 66.9 | 43.8 | 0.32 |
| Static judge                        | 71.3    | 72.5 | 53.5 | 0.24 | 68.4     | 69.6 | 48.0 | 0.29 |
| Static tools                        | 72.4    | 73.6 | 54.9 | 0.24 | 70.0     | 71.2 | 51.3 | 0.26 |
| Prompt optimization                 | 74.3    | 75.5 | 56.8 | 0.22 | 70.6     | 71.8 | 51.3 | 0.26 |
| Program search                      | 74.3    | 75.7 | 56.5 | 0.22 | 71.6     | 72.9 | 53.4 | 0.24 |
| IterEval                            | 75.8    | 77.1 | 59.9 | 0.20 | 72.2     | 73.6 | 53.4 | 0.24 |

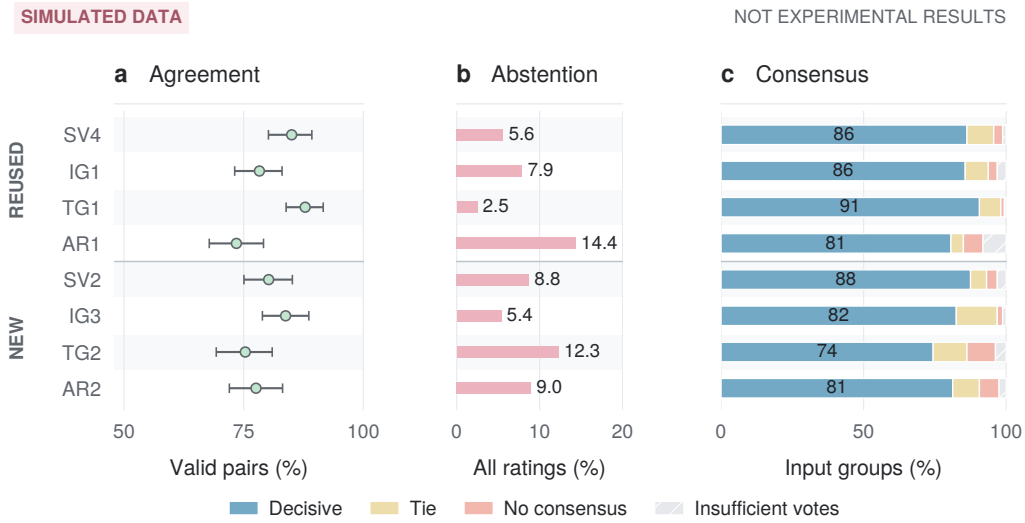
**SIMULATED DATA**

## NOT EXPERIMENTAL RESULTS



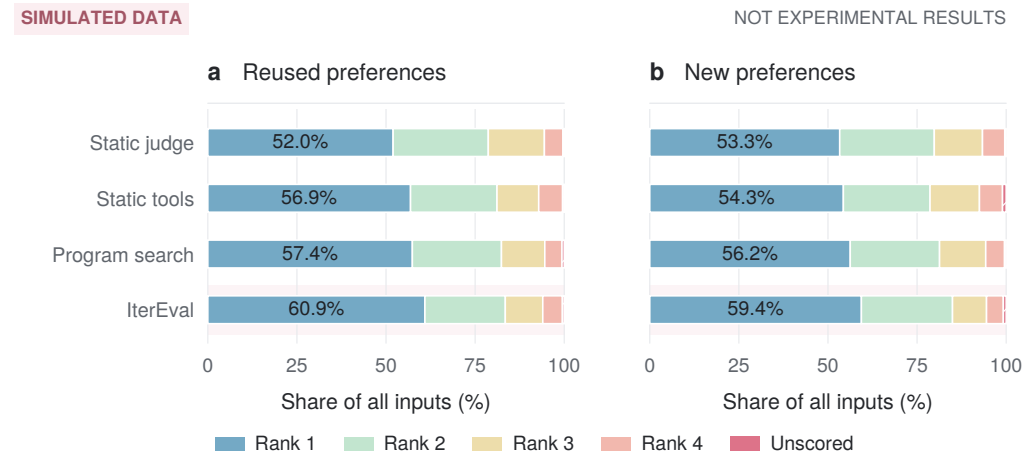
Bands: 10–90% of 12 toy runs. Each budget includes the same 10-unit random anchor.

Figure 8: **Simulated sampling curves; not experimental results.** Panels (a) and (b) compare five sampling policies under reused and newly collected preferences. Lines average 12 toy runs; bands mark each policy’s 10th–90th percentiles, not confidence intervals. Budgets count original-input annotation units, including a shared 10-unit random anchor.



Three simulated raters per group. Agreement, abstention, and consensus use distinct denominators.

Figure 9: **Simulated annotation audit; no human study was conducted.** Panel (a) averages valid-pair agreement within input groups; whiskers are 95% group-bootstrap intervals. Panel (b) divides abstentions by all requested ratings. Panel (c) retains decisive preferences, ties, no consensus, and insufficient votes under a two-of-three rule. Domain codes map to Table 13. The settings share a schema, not an assumed reliability ordering.



ID test; equal-sized toy cohorts. Rank 1 is best. Failed scoring remains in the denominator.

Figure 10: **Simulated C1 selection; not experimental results.** Panels (a) and (b) show horizontal rank profiles for four evaluator fixtures in the two construction settings, using ID inputs. Rank 1 is best. Labels give its share of all input groups, including unscored groups in the denominator. Colors encode ranks, not methods. Equal cohort sizes permit pooling within a setting; actual domain-level results must also be reported.

**Mechanisms and feedback detail.** The heatmap in Figure 11a specifies conditional contrasts between a full program and a program holding one component fixed. Such contrasts are not additive causal contributions. Panel (b) varies cumulative feedback on the same synthetic input groups. Its distributions describe assigned group-level values rather than independent search runs. Actual granularity studies must also account for annotation time and unequal label availability.

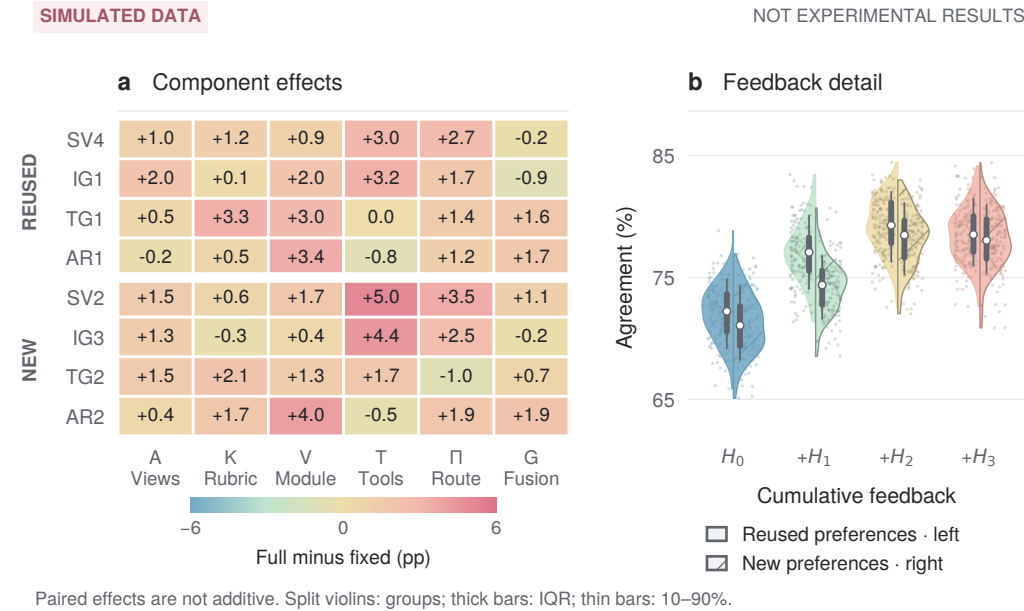
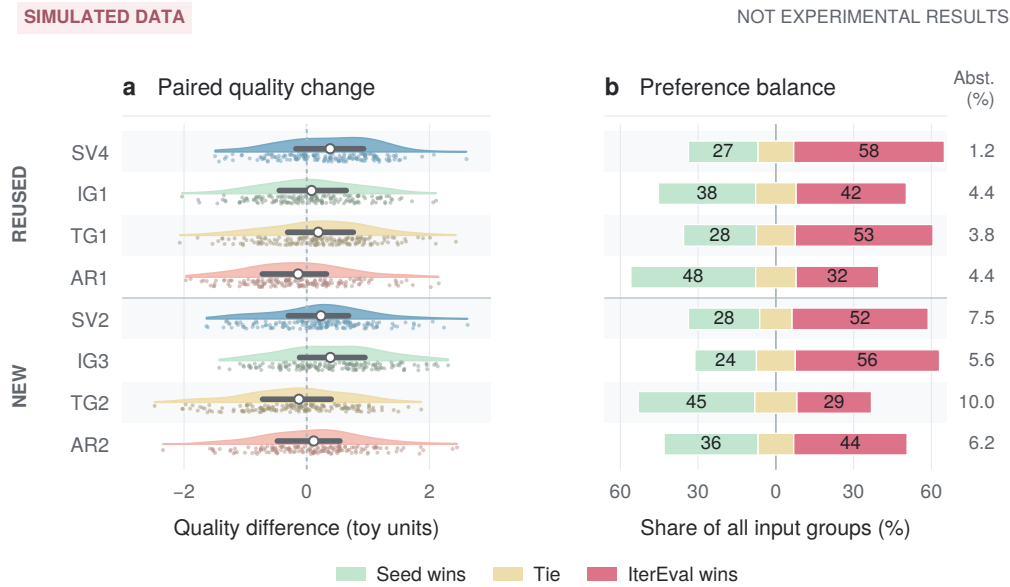


Figure 11: **Simulated ablations; not experimental results.** Panel (a) gives mean full-minus-fixed agreement (percentage points) for  $(A, K, V, T, \Pi, G)$  across eight diagnostic domains, with cream at zero. Panel (b) colors cumulative  $H_0$ – $H_3$  levels. Left/right halves denote reused/new preferences; right halves are hatched. Dots are synthetic groups, white markers medians, thick segments quartiles, and thin segments the 10th–90th percentiles. Domain codes follow Table 13. Conditional contrasts are not additive contributions or evidence of component benefits.

**Refinement and resource use.** The C2 fixture constructs two noisy refinements from a shared latent  $y_0$ . Figure 12a retains their paired quality differences and panel (b) converts those differences to wins, ties, and losses with an independent abstention indicator. The toy tie threshold is 0.15 latent units, not a proposed universal human threshold. Reused/new row labels describe feedback used for evaluator evolution; endpoint judgments on fresh C2 outputs must be new in both settings. Changes against  $y_0$  and major-regression flags are retained in the data for the full reporting contract.

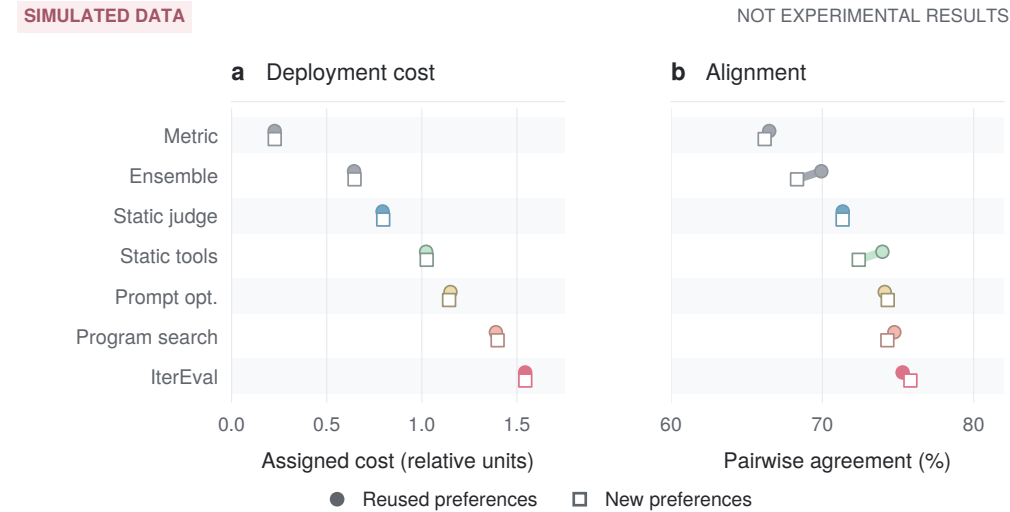
Figure 13a–b aligns assigned relative deployment costs and agreement on common evaluator rows, keeping both settings visible. Connecting segments pair settings, not steps of evolution. A real resource comparison must report search and annotation costs separately from deployment cost; the artificial costs here cannot establish practical efficiency.

**Reproduction and replacement.** The offline script `scripts/render_simulated_results.py`, seeded with 20260916, produces the CSV fixtures, vector figures, numerical table, and a manifest of hashes. Every data row carries a `SIMULATED` provenance field, and every chart discloses its status within the image. The committed assets compile without model access. Before a results-bearing submission, replace the fixtures with authorized measurements, regenerate captions and uncertainty, and audit all claims against the actual evidence. Removing a disclosure without replacing its data is not a valid conversion to results.



Difference = IterEval-refined minus Seed-refined. All artifacts and judgments here are synthetic.

Figure 12: **Simulated C2 outcomes; not experimental results.** Panel (a) combines half-violin densities, all synthetic group values, median markers, and interquartile segments for IterEval-refined minus Seed-refined quality. Density colors identify domain families. The horizontal scale is an arbitrary latent unit, not a human score. Panel (b) centers ties between seed and IterEval wins, with abstentions in a separate percentage column. Every share uses all input groups under the toy rule. Both construction settings would require fresh blinded human endpoint judgments in an actual C2 study.



Lines pair the two settings for one evaluator. Costs are assigned, not measured compute.

Figure 13: **Simulated quality-cost trade-offs; not measured costs.** Panel (a) gives assigned costs and panel (b) gives agreement, each averaging eleven ID cohorts equally per setting. Colors identify evaluators; metric-only references are gray. Filled circles denote reused preferences and hollow squares new preferences; connecting segments pair the settings for one evaluator. Cost units are assigned to the toy generator, not timed API calls or measured compute. The plot does not establish an efficiency frontier for IterEval.

## M EXPERIMENT MANIFEST

Table 15: Information required before confirmatory evaluation. Unset values are intentional placeholders, not inferred development defaults.

| Item                | Required specification                                                                                                              |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Domain and sources  | Versioned taxonomy, decision rule, readiness audit, dataset versions, permissions, input counts, output groups: <i>TBD</i>          |
| Human feedback      | Direct/derived provenance, $H_0-H_3$ availability, core dimensions, annotator pool, agreement, time, tie policy: <i>TBD</i>         |
| Partitioning        | Original-input/source grouping, deduplication, split sizes, OOD definition, External role, contamination review: <i>TBD</i>         |
| Mining              | Source inventory, mappings, dependence groups, anchor size, calibration, strata, frozen policy, human-budget accounting: <i>TBD</i> |
| Models              | Frozen scorer, proposer, tool-model, and generator identities, revisions, decoding, and permissible evidence: <i>TBD</i>            |
| Program and search  | Seed, editable components, edit bounds, allocation, development guard, protection criteria, shortlist rule: <i>TBD</i>              |
| Measurement         | Repeats $R \geq 3$ , precision, completeness, retries, artifact reuse, cache identity, search seeds, stopping rule: <i>TBD</i>      |
| Budgets             | Human time, opportunities, model calls/tokens, tools, runtime, deployment cost, training compute, matching rule: <i>TBD</i>         |
| Downstream controls | C1 pool size; C2 fresh generation and refinement; C3 reward, LoRA/GRPO settings, pilot gates and preflight: <i>TBD</i>              |
| Analysis            | Primary contrasts, grouped intervals, multiplicity, missingness, reserved metrics, minimum effects, final sample sizes: <i>TBD</i>  |

Before submission, complete Table 15, replace result placeholders with permitted source-traceable measurements, finalize the empirical scope without changing the method’s definition of a domain, and supply appropriately anonymized reproducibility artifacts. Unmeasured hypotheses remain hypotheses.